

Ben-Gurion University of the Negev
Faculty of Engineering Science
School of Electrical and Computer Engineering
Department of Electrical and Computer Engineering

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Final Report

Classification of Objects at a Very Low Resolution
Robustness of Deep Visual Models under THz-like Degradation

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Students	Bahat Itamar ID: 205805856 Dotan Ann Hanna ID: 211542311
Email	<i>bahit@post.bgu.ac.il, anndo@post.bgu.ac.il</i>
Supervisors	Prof. Yitzhak Yitzhaky Dr. Rinat Lerner
Sponsors	—
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Abstract

Background. Terahertz (THz) imaging sensors produce frames that are simultaneously low resolution, low signal-to-noise, and colorless. ImageNet-pretrained classifiers, which rely on fine textures and saturated chroma, lose accuracy quickly under those conditions, but the magnitude and the mechanism of the loss across modern deep-learning architectures are not documented quantitatively.

Aim. Compare the robustness of three representative deep architectures (a residual CNN, a densely connected CNN, and a foveal vision transformer) under controlled synthetic THz-like degradation, decompose the loss into per-axis contributions, and quantify how much of it can be recovered by training-time regularization.

Methods. The 402-cell campaign trains ResNet50, DenseNet121, and TransNeXt-tiny on CIFAR-10 and MNIST under a five-level degradation curve (saturation, downsample, blur, additive noise, salt and pepper). Seven phases isolate the effects: A clean baseline, B1 combined, C single-axis, D regularization recovery, B2 and B2-nr THz-protocol simplification, and C2 single-axis under the THz protocol. A 48-replicate multi-seed audit on the 24 L3 headline cells bounds reproducibility. Every degradation pipeline is anchored to a deterministic 256-image PSNR / SSIM probe so the cross-phase comparison sits on a physical scale.

Findings. Under Phase B1 on CIFAR-10, TransNeXt-tiny holds the highest accuracy margin over the two CNNs at the mild and moderate end of the curve (+10.1 percentage points at L1, +10.4 percentage points at L2 vs the CNN mean), and the margin narrows to under 1.5 percentage points from L3 onward. Resolution dominates the per-axis attribution by a factor of approximately four (mean L1 to L5 effect on CIFAR-10: -27.45 percentage points for resolution vs -6.58 for blur and -5.13 for salt and pepper). Regularization (T3 in Phase D) recovers approximately 0.77 percentage points of the ~ 12.5 percentage points Phase B1 accuracy collapse. The multi-seed audit confirms $\sigma < 1.5$ percentage points on every L3 headline cell.

Conclusions. The accuracy collapse is best modeled as an information bottleneck, not as an overfit. The foveal transformer holds the highest robustness margin at mild and moderate degradation; all three architectures converge into the [19, 21] percentage points band at L5. Future THz deployments should invest in resolution restoration before denoising or deblurring.

Keywords. object classification, deep learning, low resolution, THz imaging, robustness, PSNR, SSIM, calibration, transformer, residual network, dense network.

רקע. חיישני הדמיה בתחום הטרה-הרץ (THz) מפקים תמונות בעלות רזולוציה נמוכה, יחס אות-לרעש נמוך והעדר צבע. מסווגי ImageNet שאומנו מראש, המסתמכים על מרקמים עדינים ועל צבע רווי, מאבדים דיוק במהירות תחת תנאים אלה, אך עוצמת הירידה והמנגנון העומד בבסיסה אינם מתועדים כמותית עבור שלוש משפחות הארכיטקטורות המודרניות.

מטרה. להשוות את עמידותן של שלוש ארכיטקטורות-עומק מייצגות (רשת CNN שאריתית, רשת CNN צפופה, וטרנספורמר ראייה פוביאל) תחת השפלה סינתטית מבוקרת דמוית-THz, לפרק את ההפסד לתרומות לפי ציר, ולכמת כמה ממנו ניתן לשחזר על ידי רגולריזציית-אימון.

שיטות. המערכה כוללת 402 תאי אימון של ResNet50 ו-DenseNet121 ו-TransNeXt-tiny על CIFAR-10 ו-MNIST, תחת עקומת השפלה בת חמש רמות (רוויה, הקטנת רזולוציה, טשטוש, רעש גאוס, מלח-ופלפל). שבעה שלבים מבודדים את האפקטים: A קו בסיס נקי, B1 השפלה משולבת, C ציר יחיד, D שחזור באמצעות רגולריזציה, B2 ו-B2-nr לפישוט פרוטוקול ה-THz, ו-C2 ציר יחיד תחת אותו פרוטוקול. בדיקת רבייה על 48 חזרות עם זרעי-אקראיות נוספים שולחת חסם על שונות-בין-ריצות בכל אחד מ-24 תאי הכותרת ברמה L3. כל צינור השפלה מעוגן לבדיקת SSIM / PSNR דטרמיניסטית על 256 תמונות, כך שההשוואה בין השלבים מתבצעת על סקאלה פיזיקלית.

ממצאים. תחת שלב B1 ב-CIFAR-10, TransNeXt-tiny שומר על המרווח הרחב ביותר מעל שתי רשתות ה-CNN בקצה המתון של העקומה (כ-10.1% points percentage ברמה L1, כ-10.4% points percentage ברמה L2 מול הממוצע של ה-CNN-ים), ומרווח זה מצטמצם לפחות מ-1.5% points percentage החל מ-L3. ציר הרזולוציה דומיננטי בייחוס לפי ציר בקירוב פי ארבעה (אפקט ממוצע L1 עד L5 על CIFAR-10: 27.45% points percentage עבור רזולוציה לעומת 6.58% – עבור טשטוש ו-5.13% עבור מלח-ופלפל). רגולריזציה T3 בשלב D משחזרת בקירוב 0.77% points percentage מתוך קריסת הדיוק של 12.5% points percentage בשלב B1. בדיקת הרבייה מאשרת $\sigma < 1.5$ points percentage בכל תא כותרת ברמה L3.

מסקנות. קריסת הדיוק מתוארת היטב כצוואר-בקבוק של מידע ולא כהתאמת-יתר. הטרנספורמר הפוביאל שומר על מרווח העמידות הגבוה ביותר בקצה המתון של עקומת ההשפלה; שלוש הארכיטקטורות מתכנסות לרצועת [19, 21] percentage points ברמה L5. פריסות THz עתידיות צריכות להשקיע בשחזור רזולוציה לפני שלבי הסרת רעש או הסרת טשטוש. **מילות מפתח.** סיווג עצמים, למידה עמוקה, רזולוציה נמוכה, הדמיית THz, עמידות, PSNR, SSIM, כיוול, טרנספורמר, רשת שאריתית, רשת צפופה.

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1 Introduction

1.1 Motivation

Terahertz (THz) imaging is increasingly used in security screening, biomedical diagnostics, and through-fabric inspection because the millimeter-wave band penetrates clothing, paper, plastics, and other low-permittivity materials that block visible light. The physics of the band exacts a price: diffraction caps the achievable spatial resolution near the wavelength of the source, single-pixel detectors integrate the field over a comparatively large area, and the available noise figure is dominated by thermal background. The output is an image that is simultaneously low resolution, low signal-to-noise, and free of color information.

Standard deep-vision classifiers, almost all of which inherit ImageNet pretraining, are tuned for the opposite distribution. They depend on fine textures, sharp edges, and saturated chroma to populate the receptive fields of the early convolutional layers. When those signals are removed the accuracy of an off-the-shelf model collapses, and the failure mode is opaque: it is not obvious how much of the loss is caused by reduced resolution alone, how much by additive noise, and how much by the loss of color, nor whether training-time interventions (regularization, augmentation, architectural substitution) can recover any of the lost performance.

1.2 Problem statement

This project quantifies the accuracy degradation of three representative deep architectures under a controlled synthetic THz-like degradation, decomposes the loss into per-axis contributions, audits the reproducibility of the measurements across random seeds, and documents which training-side interventions recover lost accuracy. The architectures span the three families that currently anchor the image-classification literature: a residual convolutional network (ResNet50 [1]), a densely connected convolutional network with explicit feature reuse (DenseNet121 [2]), and a vision transformer with foveal attention (TransNeXt-tiny [3]).

The success criteria taken from the preliminary report are: at least 80 percent top-1 accuracy on the moderate-degradation operating point, an identifiable architectural winner that beats the baseline by a statistically meaningful margin, and inference latency compatible with sensor-side deployment. Those targets are revisited in Section 8 against the actual measurements.

1.3 Alternative approaches considered

Three alternative families were reviewed during the preliminary design and rejected in favor of synthetic-degradation training.

(i) Denoise then classify. A pipeline that restores the image before invoking a clean-image classifier. The approach decouples the two problems and reuses an existing pretrained classifier verbatim, but it requires a high-quality reference image at train time, which a real THz sensor never produces. Restoration networks trained on mismatched degradation distributions tend to hallucinate detail [4, 5], which inflates PSNR / SSIM scores without improving downstream classification.

(ii) Super-resolution then classify. A specialization of (i) that targets the resolution axis only. It assumes that the dominant failure mode is spatial sampling. As Section 5 will show, resolution does dominate the per-axis attribution, but additive noise and blur still cost between five and seven percentage points each at the moderate operating point, so a resolution-only restoration cannot recover the full loss.

(iii) Train-time degradation exposure. The model is trained directly on synthetically degraded images so that the representation adapts to the degraded distribution. This is the approach used in modern robustness benchmarks [6] and the one adopted here. It is sample-efficient (no separate restoration network), it keeps the test-time pipeline minimal (single forward pass), and it surfaces the architectural sensitivities directly because every backbone sees the same degraded distribution.

1.4 Contributions

This work contributes:

- a 402-cell training campaign across three architectures, two datasets, five degradation levels, and six experimental phases, all trained to convergence under a single fair-comparison invariant (224×224 uniform input, frozen Optuna L3 hyperparameters, identical data pipeline);
- a 48-replicate multi-seed audit on the 24 L3 headline cells that bounds run-to-run variance below 1.5 percentage points;
- a deterministic 256-image PSNR / SSIM probe per pipeline so that accuracy can be compared against the physical image-quality scale, not only against the discrete level index;
- a per-axis attribution decomposition at L3 that shows resolution is the dominant cost ($\sim 4\times$ blur or salt and pepper);
- a regularization-recovery sweep (Phase D) and a protocol simplification sweep (Phase B2 / B2-nr) that, together, quantify the fraction of the accuracy collapse that is information-bottleneck versus training noise.

1.5 Roadmap

Section 3 reports the as-built technical specification of the delivered software product and lists every drift from the preliminary specification. Section 4 documents the engineering approach and the mathematical model. Section 5 reports the final acceptance tests, including the per-pipeline image-quality probe

and the seven per-phase accuracy tables. Section 6 enumerates the problems encountered during implementation and how each was resolved. Section 8 compares the achieved performance to the preliminary specification and discusses goal-by-goal success.

2 Experimental phases at a glance

The campaign decomposes the THz-like robustness question into seven phases. Each phase opens with the pipeline it uses and closes with the research question that motivates it.

Phase A (Clean baseline). Pipeline: identity. Six cells (three models $\times$ two datasets). Establishes the upper-bound clean accuracy that every later phase is measured against.

Phase B1 (Combined degradation). Pipeline: saturation lerp $\rightarrow$ downsample $\rightarrow$ upsample to $224 \times 224 \rightarrow$ blur $\rightarrow$ additive noise $\rightarrow$ salt-and-pepper, with all five axes active at the same level. Thirty cells (three models $\times$ two datasets $\times$ five levels). Quantifies the realistic worst-case combined-degradation curve.

Phase C (Single-axis isolation). Pipeline: exactly one axis is active at level L while the other four are held at identity. One hundred fifty cells (three models $\times$ two datasets $\times$ five levels $\times$ five axes). Decomposes the Phase B1 collapse into per-axis contributions; the §5 plots use PSNR and SSIM on the x -axis to compare axes on a common image-quality scale.

Phase D (Regularization-recovery sweep). Pipeline: identical to Phase B1, plus one of three training-time regularization treatments T1, T2, T3 (architectural dropout, label mixing, kitchen sink). Ninety cells (three models $\times$ two datasets $\times$ three treatments $\times$ five levels). Asks whether the Phase B1 collapse is an overfit (regularization should recover it) or an information bottleneck (recovery should be marginal).

Phase B2 (THz-protocol simplification). Pipeline: Phase B1 minus the saturation and additive-noise axes; T3 regularization active. Thirty cells. Models the realistic THz acquisition (a single-channel low-resolution blurry frame with no chroma noise).

Phase B2-nr (No-regularization arm). Same pipeline as Phase B2, at L3 only, T3 off. Six cells. Decomposes the Phase B1 $\rightarrow$ B2 accuracy gain into a pure-protocol component and a regularization component.

Phase C2 (Single-axis isolation under the THz protocol). Same identity-elsewhere logic as Phase C, restricted to the three THz-relevant axes (resolution, blur, salt-and-pepper). Ninety cells. Per-axis attribution under deployment-realistic conditions.

Subsequent sections refer back to this overview for pipeline definitions and do not re-explain them. Table 1 is the single point of reference.

Table 1: Experimental phases at a glance. The campaign comprises 402 canonical training cells at seed 42 plus 48 multi-seed audit replicates at seeds 43 and 44. Each phase isolates a different facet of the accuracy collapse under the five-level THz-like degradation curve.

Phase	Cells	Axes active at L	Levels swept	Reg.	Primary research question
A	6	none (identity)	none	none	Upper-bound clean accuracy per (model, dataset).
B1	30	all 5 axes	L1–L5	none	Magnitude of the realistic combined-degradation collapse.
C	150	one axis (4 idle)	L1–L5	none	Per-axis decomposition of the Phase B1 collapse.
D	90	all 5 axes	L1–L5	T1 / T2 / T3	Is the Phase B1 collapse overfit or information bottleneck?
B2	30	3 axes (sat = 0, noise = 0)	L1–L5	T3	Effect of the realistic THz protocol on the combined curve.
B2-nr	6	same as B2	L3 only	none	Decomposition of B1→B2 into protocol versus regularization.
C2	90	one of {res, blur, S&P}	L1–L5	T3	Per-axis attribution under the THz protocol.
Multi-seed audit	48	all 24 L3 headline cells	L3 only	inherits cell	Run-to-run variance bound on every headline cell.

3 Final Technical Specification

This section reports the as-built specification of the delivered software product, and documents every drift from the preliminary specification [2, 3].

3.1 Product overview

The delivered product is a research-grade Python package that:

1. synthetically degrades a clean image dataset using a deterministic five-axis pipeline (saturation, down-sample, blur, additive noise, salt and pepper);
2. fine-tunes one of three pretrained deep backbones (ResNet50, DenseNet121, TransNeXt-tiny) on the degraded distribution under a unified training recipe;
3. evaluates the trained model on a held-out validation split and records validation accuracy, calibration, inference throughput, and per-pipeline PSNR / SSIM metrics;
4. aggregates results across the 402-cell campaign and the 48-replicate multi-seed audit into a single canonical JSON artifact and a 7-tab interactive dashboard.

3.2 Inputs, outputs, interfaces

Inputs. Standard image datasets (CIFAR-10 [7] and MNIST [8]), automatically downloaded by torchvision on first invocation. A YAML / dictionary configuration object selects the backbone, the degradation

phase, the level, the axis (for single-axis phases), the treatment (for Phase D), and the random seed.

Outputs. Per-cell training logs, validation metrics, best checkpoint, image-quality probe JSON. The canonical aggregator (`Final_Exp.json`, schema version 3) joins the 402 canonical cells with the 48 multi-seed replicates and exposes $\langle \text{val_acc} \rangle \pm \sigma$ pairs.

Hardware interface. The training stack runs on a single NVIDIA RTX 5070 (sm_120, Blackwell architecture) at bf16-mixed precision. A CPU-only fallback (32-true precision) is available for portability but is too slow to be useful at the campaign scale. There is no specialized hardware development; the system targets commodity GPU servers.

3.3 Final specification of the system

Models. Three backbones are supported. Table 2 summarizes the training recipe; the differential learning-rate convention (head $\times 10$ over backbone) is taken from the source papers.

Table 2: Training hyperparameters (final).

Parameter	CNNs (ResNet50, DenseNet121)	TransNeXt-tiny
Input resolution	224×224	224×224
Optimizer	AdamW [9]	AdamW
Schedule	Cosine annealing	Cosine annealing
Head learning rate	10^{-3}	5×10^{-4}
Backbone learning rate	10^{-4}	5×10^{-5}
Weight decay	10^{-4}	5×10^{-2}
Label smoothing	0.1	0.1
Stochastic depth	0.0	0.1
Gradient clip	$\ \nabla\ _2 \leq 1.0$	$\ \nabla\ _2 \leq 1.0$
Batch size	32	32
Max epochs	60	60
Early stopping	patience 10	patience 10
Precision	bf16-mixed	bf16-mixed
Seed	42 (canonical)	42 (canonical)

Degradation pipeline. The five-level schedule (Table 3) is the single source of truth for L1 through L5; it is imported from `src/data/degradation_levels.py` into both the training and the evaluation paths.

Data pipeline. Pipeline version 2 (finalized 2026-05-20): the saturation lerp is applied first, then the bilinear downsample to `low_res`, then additive Gaussian noise and salt and pepper at the low resolution, then bicubic upsample to 224×224 , then Gaussian blur (kernel calibrated for 224×224), then a final clip to $[0, 1]$ and ImageNet normalization.

Determinism. The per-sample degradation seed is `idx + 107` on the validation split and `idx` on the training split. The byte-identity gate in `src/tests/test_degradation_determinism.py` asserts that

Table 3: Five-level synthetic degradation schedule. Blur kernel and σ are pixel-domain values at the 224×224 model input.

Level	Name	low_res	Blur K	Blur σ	Noise std	S&P	Saturation
L1	Mild	18	13	2.50	0.04	0.03	0.95
L2	Light	12	25	5.00	0.08	0.06	0.65
L3	Moderate	8	41	8.00	0.12	0.10	0.40
L4	Severe	6	61	12.00	0.16	0.14	0.15
L5	Extreme	3	91	18.00	0.22	0.18	0.00

two independent reads of the same validation index yield bit-identical pixels across freshly built datasets; this is also verified across seeds 42, 43, and 44 by the multi-seed audit test.

3.4 Drift from the preliminary specification

The Preliminary Design Review (PDR, Pre-2026-061.pdf) defined the project scope. Five drifts occurred during implementation:

1. **TransNeXt variant: small to tiny.** The preliminary spec proposed `transnext_small` ($\sim 50\text{M}$ parameters); the 2026-05-14 revision substituted `transnext_tiny` ($\sim 28\text{M}$ parameters) to keep the parameter count comparable to ResNet50 ($\sim 25\text{M}$) and to recover ~ 310 GPU-hours across the TransNeXt rows. The change preserves the foveal-attention mechanism, which was the architectural property of interest, and it matches the published TransNeXt 224×224 checkpoint exactly.
2. **Uniform 224×224 protocol.** A short-lived V3 refactor (2026-05-08 through 2026-05-13) attempted native-resolution training (32×32 for TransNeXt on CIFAR-10, $28 \rightarrow 32$ for MNIST). The refactor violated the fair-comparison invariant (different backbones saw different tensor shapes), so it was rolled back. Every cell now trains at 224×224 .
3. **Blur kernel rescaling.** The 2026-05-12 degradation curve used blur kernels designed for native 32×32 imagery ($K = 3 \dots 13$, $\sigma = 0.8 \dots 2.3$). At 224×224 those kernels covered ~ 1 to 6% of the image width and produced essentially invisible blur. They were rescaled on 2026-05-14 to pixel-domain values at 224×224 ($K = 13 \dots 91$, $\sigma = 2.5 \dots 18$), with the kernel-to- σ rule $K = 2\lceil 2.5\sigma \rceil + 1$ so the kernel always captures the 2.5σ Gaussian footprint.
4. **Optuna freeze.** The preliminary plan reserved a per-(phase, cell) hyperparameter sweep. The final spec freezes the Optuna L3 winner per (model, dataset) and reuses it across every level and every axis. The change saves ~ 200 GPU-hours and removes a per-phase confound from the cross-phase comparison.
5. **Multi-seed audit.** The preliminary spec used a single seed (42). The 2026-06-01 revision added a 48-replicate audit at seeds 43 and 44 across the 24 L3 headline cells so that the published L3 accuracies carry an explicit σ bound rather than a single point estimate.

3.5 Phases of the experimental program

The 402-cell campaign is partitioned into seven phases. Table 4 summarizes each phase.

Table 4: Experimental phases of the 402-cell campaign. Multi-seed audit replicates are layered on top of the 24 L3 headline cells.

Phase	Purpose	Cells	Tag prefix
A	Clean baseline (no degradation)	6	final_clean_
B	Combined degradation, all axes at level L	30	final_B_L*
C	Single-axis isolation, others at identity	150	final_C_L*_<axis>_
D	Regularization recovery (T1 / T2 / T3) on Phase B	90	final_D_T*_L*_
B2	THz protocol simplification (sat = 0, noise = 0, T3)	30	final_B2_L*_
B2nr	B2 without regularization, L3 only	6	final_B2nr_L3_
C2	Single-axis under THz protocol	90	final_C2_L*_<axis>_
Audit	Multi-seed audit (seeds {43, 44}) on the 24 L3 headline cells	48	*_seed{43,44}
Total		450	

4 Engineering Design

This section documents the chosen approach in detail. The system block diagram is shown in Figure 1; the degradation pipeline that drives the experimental program is shown in Figure 2.

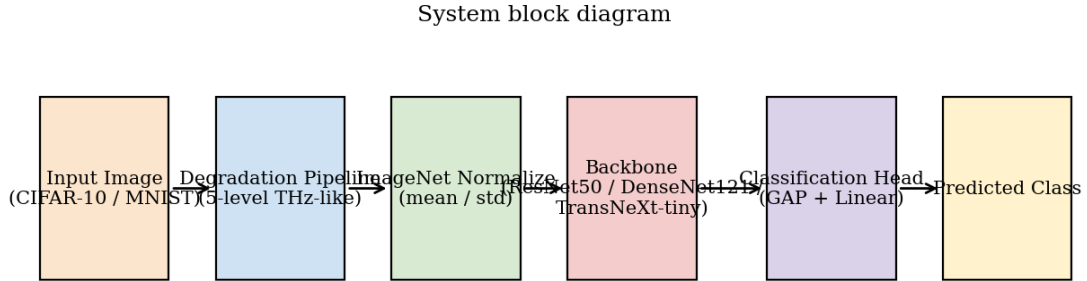


Figure 1: System block diagram. An input image from CIFAR-10 or MNIST passes through the THz-like degradation pipeline, is normalized to the ImageNet distribution, is fed to one of the three backbones, and is classified by a global-average-pooling head.

4.1 Mathematical model

Degradation operator. Let $x \in [0, 1]^{3 \times H \times W}$ be a clean image and let $\phi = (s, r, \sigma_n, K_b, \sigma_b, p)$ denote the degradation parameters: saturation $s \in [0, 1]$, low-resolution target $r \in \mathbb{N}$, noise standard deviation $\sigma_n \geq 0$, blur kernel size K_b , blur sigma σ_b , and salt-and-pepper amount $p \in [0, 1]$. Define the composite

THz-like degradation pipeline (pipeline v2)

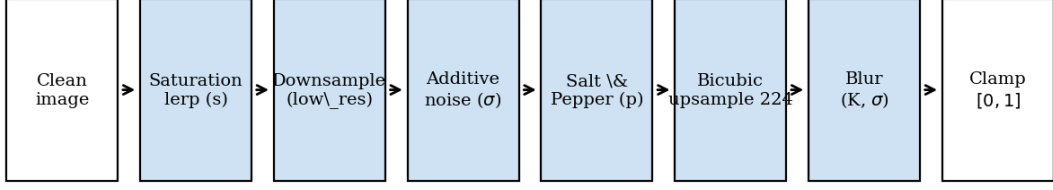


Figure 2: Pipeline version 2. The saturation lerp runs first so the grayscale luminance is computed from the un-degraded source. Additive noise and salt-and-pepper operate at the pre-upsample resolution; bicubic upsample spreads each stochastic sample over the kernel footprint, producing sensor-grain noise structure rather than fine pixel-level grain.

operator

$$\mathcal{D}_\phi(x) = \text{Clip}_{[0,1]} \circ \text{Blur}_{K_b, \sigma_b} \circ \text{Up}_{224} \circ \text{SP}_p \circ \text{N}_{\sigma_n} \circ \text{Down}_r \circ \text{Sat}_s, \quad (1)$$

where $\text{Sat}_s(x) = (1 - s) \text{Gray}(x) + s x$ with $\text{Gray}(x) = 0.2989 x_R + 0.5870 x_G + 0.1140 x_B$ (BT.601 luma); Down_r is bilinear downsample to $r \times r$; N_{σ_n} adds independent Gaussian noise of standard deviation σ_n ; SP_p flips a $p/2$ fraction of pixels to 1 and another $p/2$ fraction to 0; Up_{224} is bicubic upsample to 224×224 ; and $\text{Blur}_{K_b, \sigma_b}$ is a separable Gaussian blur with kernel size K_b and standard deviation σ_b .

Classifier. A backbone $f_\theta(\cdot)$ produces a logit vector $z = f_\theta(\mathcal{D}_\phi(x))$. With label smoothing $\alpha = 0.1$, the training objective is the smoothed cross-entropy

$$\mathcal{L}(\theta) = - \sum_{c=1}^C \tilde{y}_c \log \text{softmax}(z)_c, \quad \tilde{y}_c = (1 - \alpha) \mathbf{1}[c = y] + \frac{\alpha}{C}. \quad (2)$$

Backbones. The three architectures expose different inductive biases for the low-resolution regime. ResNet50 [1] combines features via additive identity skips:

$$y_\ell = F_\ell(x_\ell; \theta_\ell) + x_\ell. \quad (3)$$

The summation reuses previous-layer features but collapses them into the running representation; subtle low-level cues can be washed out when the later residual blocks dominate.

DenseNet121 [2] concatenates the feature maps of every preceding layer in the same dense block:

$$x_\ell = H_\ell([x_0, x_1, \dots, x_{\ell-1}]), \quad (4)$$

where H_ℓ is a BatchNorm-ReLU-Conv $_{3 \times 3}$ composite. The collective knowledge of every preceding layer is explicitly preserved, which is the property the preliminary spec hypothesized would help under

information loss.

TransNeXt-tiny [3] uses pixel-focused attention with foveal and peripheral kernels:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M_{\text{focal}}\right) V, \quad (5)$$

where M_{focal} is a window-shaped mask that down-weights all but the nearest neighbors of each query, while a parallel global pooling stream contributes a coarse-grained peripheral signal. The biomimetic prior is designed to retain accuracy when the global shape is corrupted but local texture is partially preserved.

4.2 Image-quality metrics

Each degradation pipeline is anchored to a PSNR and SSIM probe so that cross-phase comparisons sit on a physical scale.

PSNR [4] between a reference image x and a degraded image $\tilde{x} = \mathcal{D}_\phi(x)$, both in $[0, 1]^{3 \times 224 \times 224}$, is

$$\text{PSNR}(x, \tilde{x}) = 10 \log_{10} \frac{1}{\text{MSE}(x, \tilde{x})}, \quad \text{MSE}(x, \tilde{x}) = \frac{1}{3 \cdot 224^2} \sum_{c,i,j} (x_{c,i,j} - \tilde{x}_{c,i,j})^2. \quad (6)$$

SSIM [5] is the structural similarity index. For two local patches u, v with means μ_u, μ_v , variances σ_u^2, σ_v^2 , and covariance σ_{uv} ,

$$\text{SSIM}(u, v) = \frac{(2\mu_u\mu_v + C_1)(2\sigma_{uv} + C_2)}{(\mu_u^2 + \mu_v^2 + C_1)(\sigma_u^2 + \sigma_v^2 + C_2)}, \quad (7)$$

with C_1, C_2 small stabilizers; the full-image score is the spatial mean over an 11×11 Gaussian window. The implementation uses `torchmetrics.image.StructuralSimilarityIndexMeasure` with `data_range = 1`.

Reference construction. The PSNR / SSIM reference is the input image upsampled to 224×224 by bilinear interpolation. The choice matches what a hypothetical clean-image classifier would see if the degradation pipeline were short-circuited; it does not include the saturation lerp, so the measurements isolate the spatial loss from the color loss. Both choices are documented in the figure captions in Section 5 so the reader can interpret the absolute scale correctly.

Probe protocol. For each pipeline the probe iterates over the first 256 validation indices, applies the pipeline with seed offset 2×10^7 (chosen disjoint from the training and validation streams), and aggregates the mean and standard deviation. The 256-sample size balances probe runtime against the standard error of the mean (below 0.1 dB on the L3 pipelines).

4.3 Optimizer and learning-rate schedule

All cells train with AdamW [10] under decoupled weight decay. With $g_t = \nabla_{\theta} \mathcal{L}(\theta_{t-1})$ the loss gradient at step t , the running first and second moment estimates are

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad (8)$$

and the parameter update is

$$\theta_t = \theta_{t-1} - \eta_t \left(\frac{m_t / (1 - \beta_1^t)}{\sqrt{v_t / (1 - \beta_2^t)} + \epsilon} + \lambda \theta_{t-1} \right), \quad (9)$$

with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, and weight decay $\lambda \in \{10^{-4}, 5 \times 10^{-2}\}$ (CNN / TransNeXt). Gradient clipping caps the global norm at 1.0.

The learning rate η_t follows a cosine annealing schedule [11] between $\eta_{\max}$ and $\eta_{\min}$ over a total budget of T training steps:

$$\eta_t = \eta_{\min} + \frac{1}{2} (\eta_{\max} - \eta_{\min}) \left(1 + \cos \left(\frac{t}{T} \pi \right) \right). \quad (10)$$

Differential learning rates apply $\eta_{\max}$ separately to the classifier head (10^{-3} on CNNs, 5×10^{-4} on TransNeXt-tiny) and to the backbone parameters (10^{-4} on CNNs, 5×10^{-5} on TransNeXt-tiny). Training uses bf16-mixed precision on the RTX 5070 host and runs for at most 60 epochs with early stopping at patience 10 on validation accuracy.

4.4 Calibration metric

Expected calibration error (ECE) [12, 13] bins the predicted probabilities into $M = 15$ equal-width bins and computes

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (11)$$

where B_m is the set of predictions whose top-1 probability falls in bin m , $\text{acc}(B_m)$ is the empirical accuracy of B_m , and $\text{conf}(B_m)$ is the mean predicted probability in B_m . A well-calibrated model has $\text{ECE} \rightarrow 0$.

4.5 Regularization treatments (Phase D)

Phase D layers three regularization treatments on top of the frozen Phase B Optuna winners:

T1. Dropout (CNN, rate 0.2) or stochastic-depth drop-path [14] (TransNeXt, rate 0.2). Architectural regularization only; no label-side intervention.

T2. mixup [15] with $\alpha = 0.2$; label-mixing only.

T3. The combination: dropout (or drop-path) + mixup $\alpha = 0.2$ + CutMix [16] $\alpha = 1.0$. The kitchen-sink combination defines the upper bound of training-side recovery measurable inside the design space.

4.6 Software architecture

The implementation is layered:

- `src/data/` hosts the single source of truth for the degradation curve (`degradation_levels.py`) and the operator (`degrade.py`). Both modules are imported verbatim into the training and the image-quality probe paths, which is the property that guarantees the measured PSNR / SSIM correspond to the pixels the model actually sees.
- `src/lightning/` hosts the training engine (`THzClassifier`, `THzDataModule`) built on PyTorch Lightning [17].
- `src/models/` hosts the backbone wrappers; the TransNeXt weights are auto-downloaded from the public checkpoint and cached under `artifacts/weights/`.
- `scripts/` hosts the campaign drivers (`run_all_phases.py`) and the diagnostics builders.
- `Final_Report/scripts/` hosts the figure / table / sample generators that produce this report. Every artifact under `Final_Report/` is reproducible by running `python Final_Report/scripts/build_*.py` from a clean checkout.

5 Final Acceptance Tests

This section reports the measured performance of the delivered product against every relevant metric. Every value comes from a frozen training run recorded under `runs/final/` and aggregated by `artifacts/Final_Exp.json` (schema version 3). No values are synthesized or interpolated.

5.1 Test rig

The campaign ran on a single workstation with the configuration in Table 5. Determinism is guarded by `pl.seed_everything(42, workers=True)` for the canonical seed and by the per-sample seed offset for the validation split.

5.2 Per-pipeline image quality

Table 6 reports the mean PSNR and SSIM for every degradation pipeline that appears in the experiments. The values are identical across same-pipeline cells by construction, so Phase D rows inherit the Phase B

Table 5: Test rig and software stack.

Component	Value
GPU	NVIDIA RTX 5070 (Blackwell, sm_120, 12 GB)
Precision	bf16-mixed
CPU dataloader workers	0 (Windows)
Framework	PyTorch 2.11 [18] + Lightning 2.x [17]
Image quality	torchmetrics PSNR / SSIM, data_range = 1
Determinism gate	src/tests/test_degradation_determinism.py

PSNR / SSIM at the same level. Figures 3 and 4 show the visual appearance of the pipelines on CIFAR-10; Figures 5 and 6 repeat the comparison on MNIST.

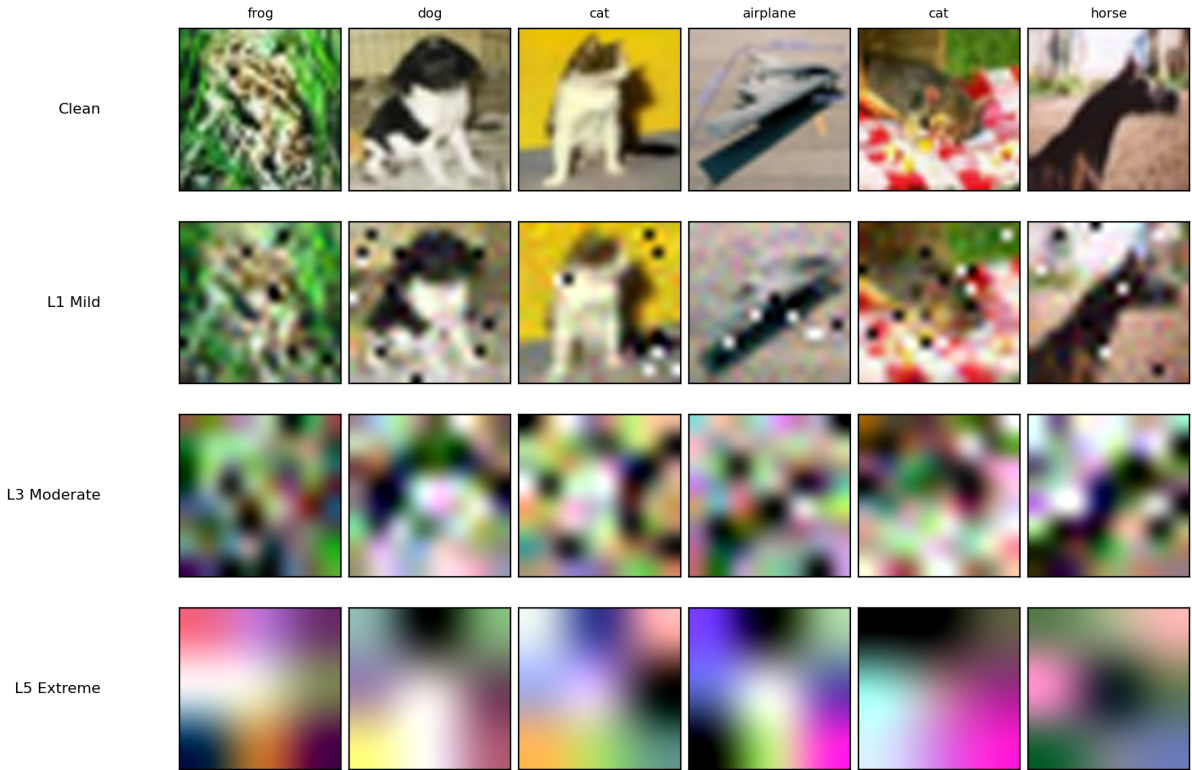


Figure 3: CIFAR-10: six fixed validation examples (deterministic seed) at clean, L1 Mild, L3 Moderate, and L5 Extreme combined degradation. Class labels above each column. Rows from top to bottom: Clean, L1, L3, L5.

5.3 Clean baseline accuracy (Phase A)

Table 7 reports the validation accuracy at the clean operating point. These values are the upper bounds against which every subsequent phase is measured.

Table 6: Mean PSNR and SSIM per degradation pipeline. Each value is computed over a fixed deterministic 256-sample validation subset using torchmetrics with data_range=1, sample seed offset = 2×10^7 , clean reference upsampled to 224×224 by bilinear interpolation.

Dataset	Phase	Level	Axis	PSNR (dB)	SSIM
CIFAR-10	B	L1	–	21.23 ± 1.54	0.801 ± 0.041
CIFAR-10	B	L2	–	17.54 ± 1.32	0.697 ± 0.059
CIFAR-10	B	L3	–	14.87 ± 1.41	0.635 ± 0.078
CIFAR-10	B	L4	–	13.10 ± 1.41	0.601 ± 0.087
CIFAR-10	B	L5	–	10.85 ± 1.81	0.566 ± 0.110
CIFAR-10	B2	L1	–	19.50 ± 2.31	0.816 ± 0.054
CIFAR-10	B2	L2	–	17.25 ± 1.88	0.733 ± 0.067
CIFAR-10	B2	L3	–	15.24 ± 1.72	0.673 ± 0.079
CIFAR-10	B2	L4	–	14.18 ± 1.86	0.652 ± 0.086
CIFAR-10	B2	L5	–	12.40 ± 2.50	0.624 ± 0.108
CIFAR-10	B2nr	L3	–	15.24 ± 1.72	0.673 ± 0.079
CIFAR-10	C	L1	Blur	41.98 ± 2.54	0.983 ± 0.007
CIFAR-10	C	L1	Noise	28.60 ± 0.42	0.841 ± 0.021
CIFAR-10	C	L1	Resolution	30.78 ± 2.70	0.917 ± 0.035
CIFAR-10	C	L1	S&P	21.66 ± 1.10	0.858 ± 0.023
CIFAR-10	C	L1	Saturation	36.45 ± 2.39	0.975 ± 0.010
CIFAR-10	C	L2	Blur	33.58 ± 2.59	0.952 ± 0.020
CIFAR-10	C	L2	Noise	23.34 ± 0.36	0.642 ± 0.038
CIFAR-10	C	L2	Resolution	25.33 ± 2.65	0.845 ± 0.060
CIFAR-10	C	L2	S&P	18.61 ± 0.87	0.755 ± 0.032
CIFAR-10	C	L2	Saturation	31.34 ± 3.42	0.968 ± 0.014
CIFAR-10	C	L3	Blur	27.22 ± 2.54	0.885 ± 0.045
CIFAR-10	C	L3	Noise	20.12 ± 0.42	0.489 ± 0.047
CIFAR-10	C	L3	Resolution	21.75 ± 2.64	0.784 ± 0.079
CIFAR-10	C	L3	S&P	16.38 ± 0.70	0.635 ± 0.035
CIFAR-10	C	L3	Saturation	27.94 ± 4.23	0.958 ± 0.022
CIFAR-10	C	L4	Blur	23.63 ± 2.52	0.822 ± 0.066
CIFAR-10	C	L4	Noise	17.87 ± 0.44	0.381 ± 0.050
CIFAR-10	C	L4	Resolution	19.42 ± 2.67	0.745 ± 0.091
CIFAR-10	C	L4	S&P	14.96 ± 0.60	0.538 ± 0.038
CIFAR-10	C	L4	Saturation	25.41 ± 4.72	0.945 ± 0.033
CIFAR-10	C	L5	Blur	21.06 ± 2.52	0.772 ± 0.082
CIFAR-10	C	L5	Noise	15.49 ± 0.43	0.277 ± 0.048
CIFAR-10	C	L5	Resolution	15.88 ± 2.82	0.692 ± 0.109
CIFAR-10	C	L5	S&P	13.88 ± 0.58	0.456 ± 0.041
CIFAR-10	C	L5	Saturation	24.17 ± 4.93	0.936 ± 0.041
CIFAR-10	C2	L1	Blur	24.67 ± 5.66	0.943 ± 0.040
CIFAR-10	C2	L1	Resolution	23.11 ± 3.97	0.883 ± 0.050
CIFAR-10	C2	L1	S&P	19.27 ± 2.05	0.824 ± 0.043
CIFAR-10	C2	L2	Blur	23.74 ± 4.42	0.915 ± 0.044
CIFAR-10	C2	L2	Resolution	21.22 ± 3.09	0.819 ± 0.068
CIFAR-10	C2	L2	S&P	17.19 ± 1.43	0.724 ± 0.044
CIFAR-10	C2	L3	Blur	22.05 ± 3.35	0.855 ± 0.057
CIFAR-10	C2	L3	Resolution	19.34 ± 2.65	0.763 ± 0.082
CIFAR-10	C2	L3	S&P	15.46 ± 1.03	0.609 ± 0.044
CIFAR-10	C2	L4	Blur	20.43 ± 2.77	0.798 ± 0.071
CIFAR-10	C2	L4	Resolution	17.79 ± 2.47	0.729 ± 0.092

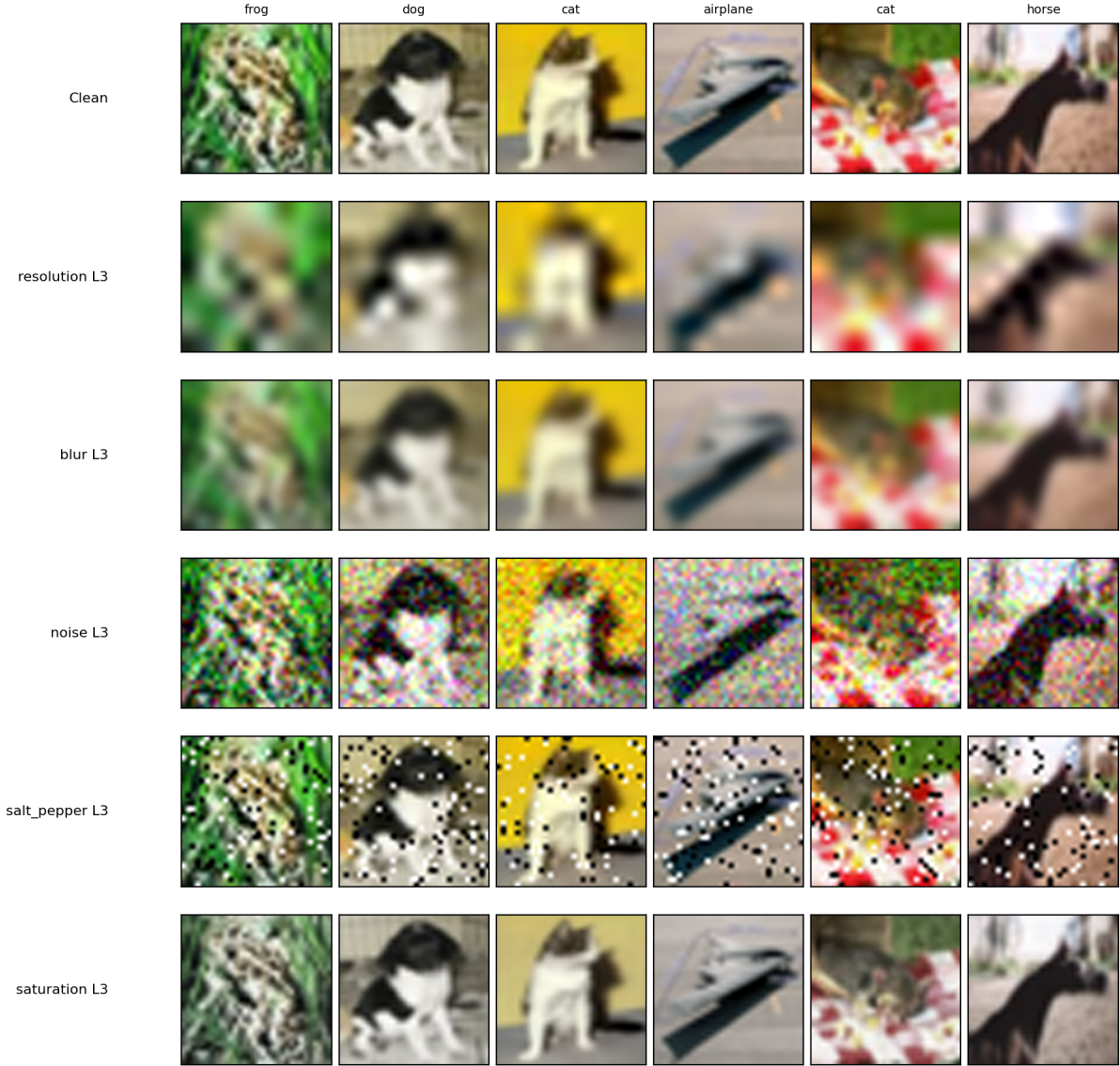


Figure 4: CIFAR-10: per-axis isolation at L3 (Phase C). Each row shows the same six images with only the named axis active.

Table 7: Phase A clean-baseline accuracy (%) on the held-out validation split. No degradation applied; values are upper bounds for every subsequent phase.

Model	CIFAR-10	MNIST
ResNet50	95.18	99.14
DenseNet121	93.56	99.30
TransNeXt-tiny	97.64	99.24

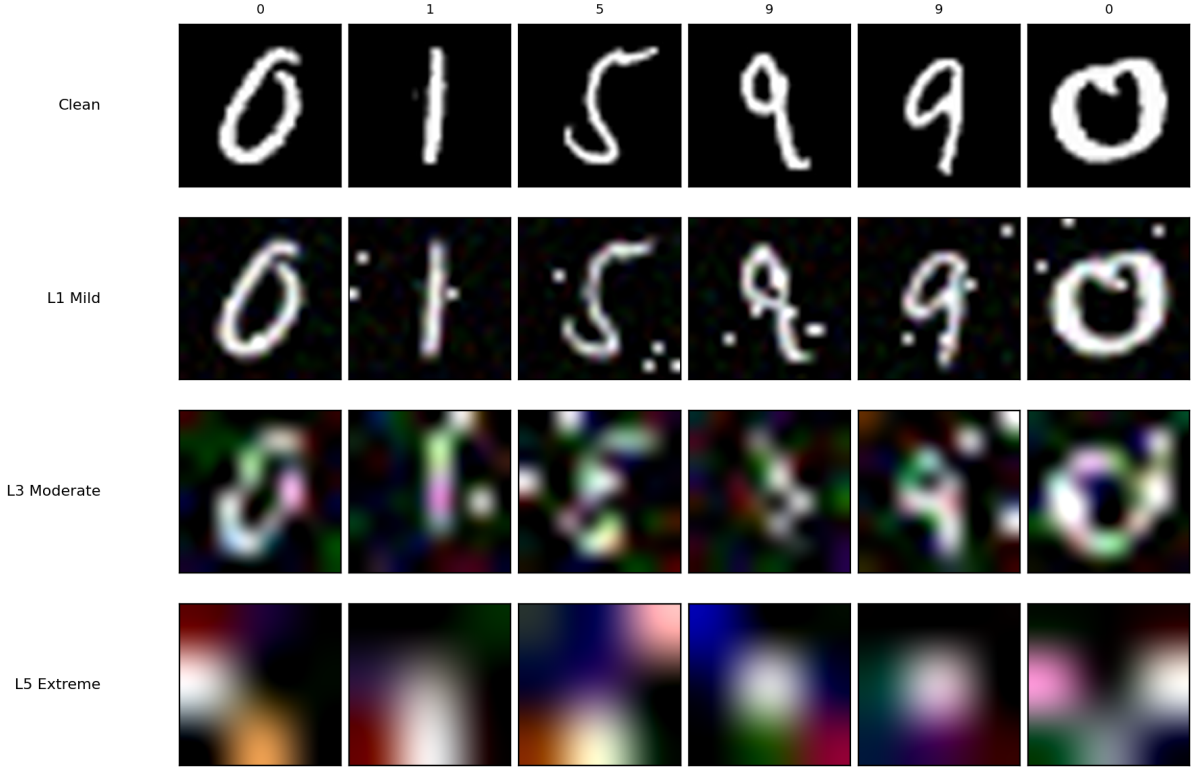


Figure 5: MNIST: combined degradation at L1, L3, L5 on six fixed validation examples.

5.4 Combined-degradation accuracy (Phase B)

Phase B applies every axis simultaneously at the same level. Table 8 reports the full $3 \times 2 \times 5$ accuracy block; Figure 7 plots the curves with the Phase D-T3 recovery overlay.

Table 8: Phase B (combined degradation) validation accuracy (%). All five axes active at the same level L. Cells with $\pm$ values are multi-seed audited (seeds {42, 43, 44}).

Model	Dataset	L1	L2	L3	L4	L5
ResNet50	CIFAR-10	76.46	53.94	36.31 ± 0.83	26.84	19.06
ResNet50	MNIST	99.18	96.36	79.17 ± 0.80	56.44	28.00
DenseNet121	CIFAR-10	78.48	53.68	38.54 ± 0.78	28.52	19.58
DenseNet121	MNIST	98.74	96.54	80.79 ± 0.84	58.08	27.76
TransNeXt-tiny	CIFAR-10	87.58	64.22	39.48 ± 0.62	28.50	20.10
TransNeXt-tiny	MNIST	98.96	96.00	80.05 ± 0.11	57.24	27.48

5.5 Accuracy versus image quality

Figures 8 and 9 plot validation accuracy against the per-pipeline mean PSNR and SSIM respectively. The Spearman rank correlation is reported in each panel and is monotonically strong on both metrics for both datasets. Accuracy and image quality collapse together, but the slope is not unity: on CIFAR-10 the model loses accuracy faster than SSIM, while on MNIST the two metrics track each other closely. This

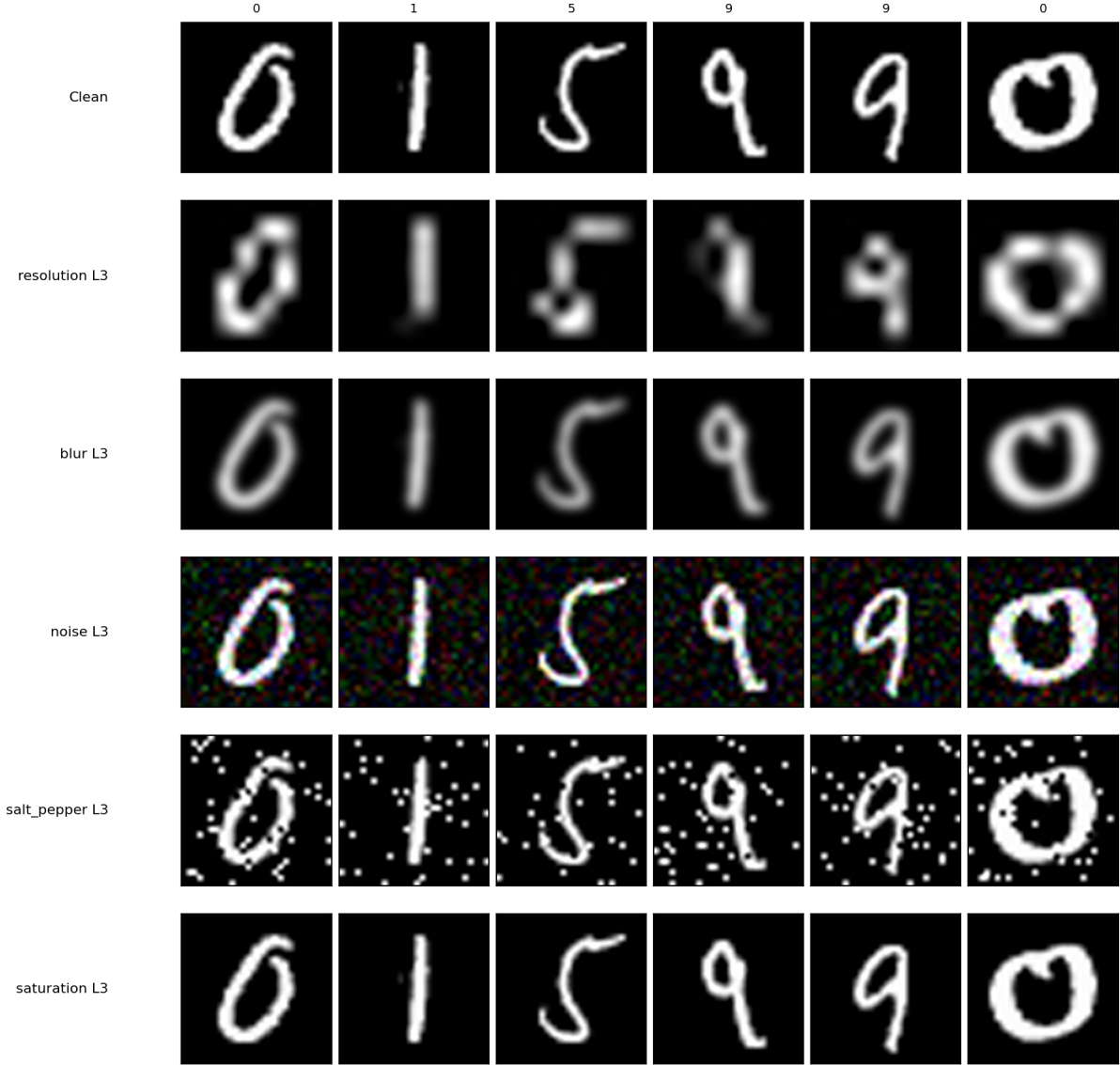


Figure 6: MNIST: per-axis isolation at L3 (Phase C).

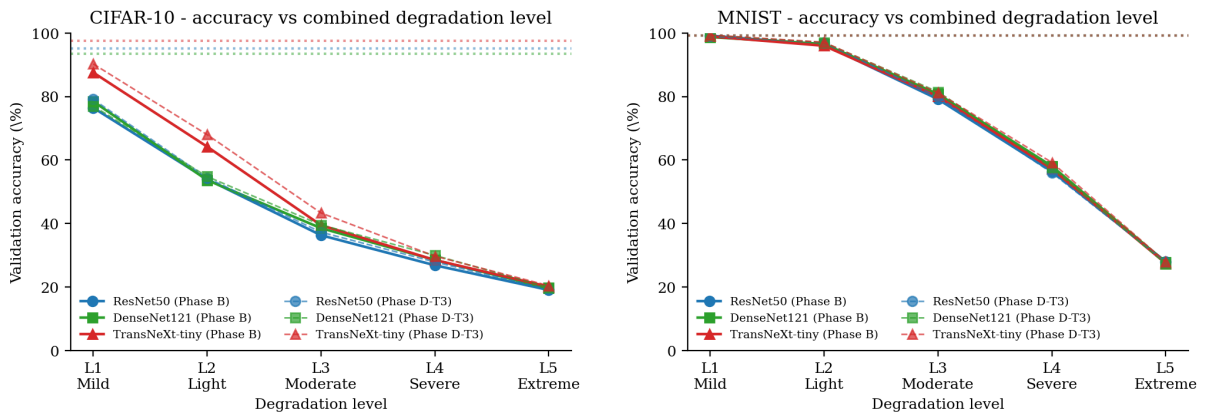


Figure 7: Validation accuracy vs combined degradation level. Solid curves: Phase B (no extra regularization). Dashed curves: Phase D-T3 (T3 regularization on the same pipeline). Dotted horizontal lines: Phase A clean baseline per model.

corroborates the per-axis attribution in Section 5.6: CIFAR-10 carries more inter-class color information that is destroyed by the saturation lerp, and SSIM does not penalize the color collapse strongly.

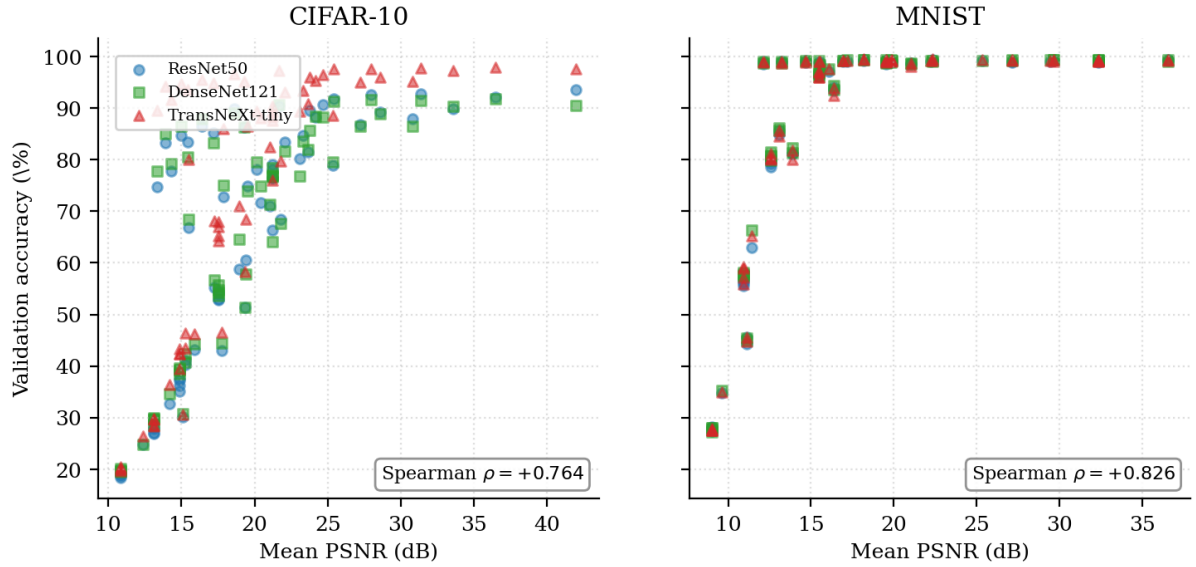


Figure 8: Validation accuracy vs mean PSNR per pipeline. Each marker is one trained cell; color indicates the backbone. The strong negative monotone relationship justifies treating PSNR as a low-cost proxy for downstream accuracy at design time.

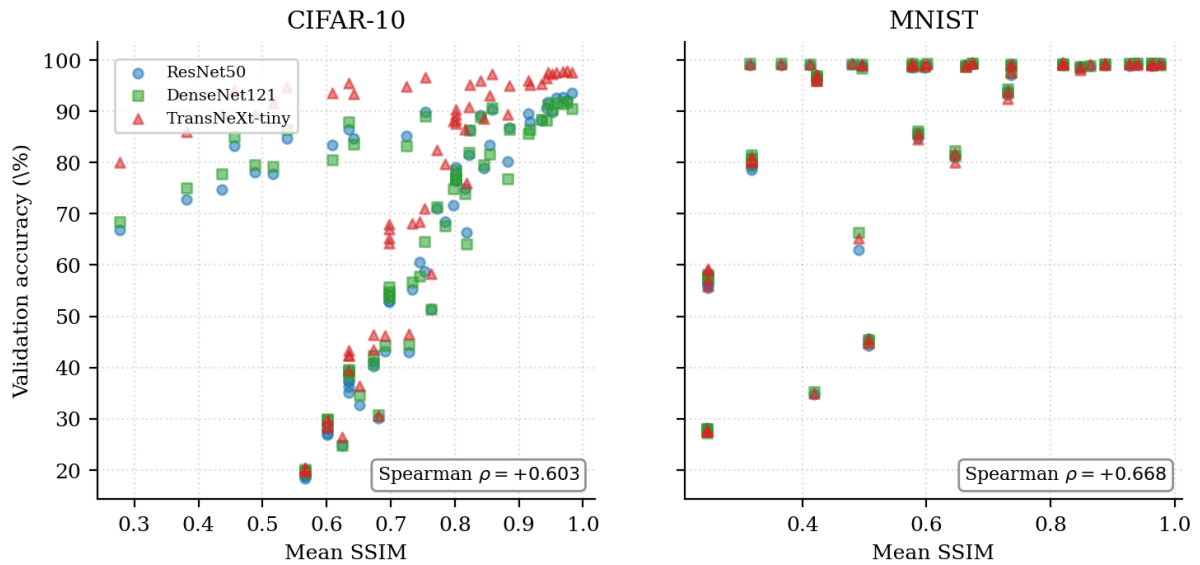


Figure 9: Validation accuracy vs mean SSIM per pipeline.

5.6 Per-axis isolation (Phase C)

Phase C isolates each axis: the named axis is at level L , every other axis is at its identity value. Table 9 reports the full $3 \times 2 \times 5 \times 5 = 150$ accuracy block; Figure 10 shows the per-axis attribution at L3 as accuracy delta vs the Phase A clean baseline.

Table 9: Phase C single-axis isolation validation accuracy (%). The named axis is at level L; every other axis is at its identity value.

Model	Dataset	Axis	L1	L2	L3	L4	L5
ResNet50	CIFAR-10	Resolution	88.00	78.98	68.46	60.50	43.16
ResNet50	CIFAR-10	Blur	93.54	89.88	86.78	81.46	70.98
ResNet50	CIFAR-10	Noise	89.14	84.70	78.06	72.76	66.88
ResNet50	CIFAR-10	S&P	90.42	89.92	86.52	84.64	83.30
ResNet50	CIFAR-10	Saturation	92.06	92.82	92.56	91.78	88.22
ResNet50	MNIST	Resolution	98.84	98.50	93.48	81.22	44.26
ResNet50	MNIST	Blur	99.32	99.14	99.18	99.20	98.82
ResNet50	MNIST	Noise	99.10	99.16	99.10	99.10	98.96
ResNet50	MNIST	S&P	99.08	98.98	98.88	98.90	98.62
ResNet50	MNIST	Saturation	98.94	99.18	99.36	99.18	99.10
DenseNet121	CIFAR-10	Resolution	86.52	79.60	67.68	57.76	44.32
DenseNet121	CIFAR-10	Blur	90.54	90.40	86.44	82.04	71.42
DenseNet121	CIFAR-10	Noise	88.92	83.64	79.54	75.06	68.46
DenseNet121	CIFAR-10	S&P	90.72	88.98	87.96	86.44	85.08
DenseNet121	CIFAR-10	Saturation	91.84	91.42	91.66	91.30	88.34
DenseNet121	MNIST	Resolution	99.22	98.62	93.60	81.32	45.40
DenseNet121	MNIST	Blur	99.06	99.34	98.96	99.28	98.46
DenseNet121	MNIST	Noise	99.34	99.26	99.24	99.34	99.28
DenseNet121	MNIST	S&P	99.18	99.14	99.06	98.76	98.72
DenseNet121	MNIST	Saturation	99.32	99.24	99.26	99.28	99.14
TransNeXt-tiny	CIFAR-10	Resolution	95.14	88.62	79.64	68.40	46.24
TransNeXt-tiny	CIFAR-10	Blur	97.66	97.24	94.96	90.88	82.44
TransNeXt-tiny	CIFAR-10	Noise	95.98	93.34	89.40	85.98	80.08
TransNeXt-tiny	CIFAR-10	S&P	97.28	96.60	95.48	94.74	94.22
TransNeXt-tiny	CIFAR-10	Saturation	97.92	97.78	97.54	97.56	95.34
TransNeXt-tiny	MNIST	Resolution	99.14	98.08	92.48	80.08	44.86
TransNeXt-tiny	MNIST	Blur	99.24	98.96	99.14	99.28	99.06
TransNeXt-tiny	MNIST	Noise	99.16	99.28	99.24	99.16	99.14
TransNeXt-tiny	MNIST	S&P	99.00	99.16	98.82	98.76	98.94
TransNeXt-tiny	MNIST	Saturation	99.36	98.96	99.18	99.24	99.04

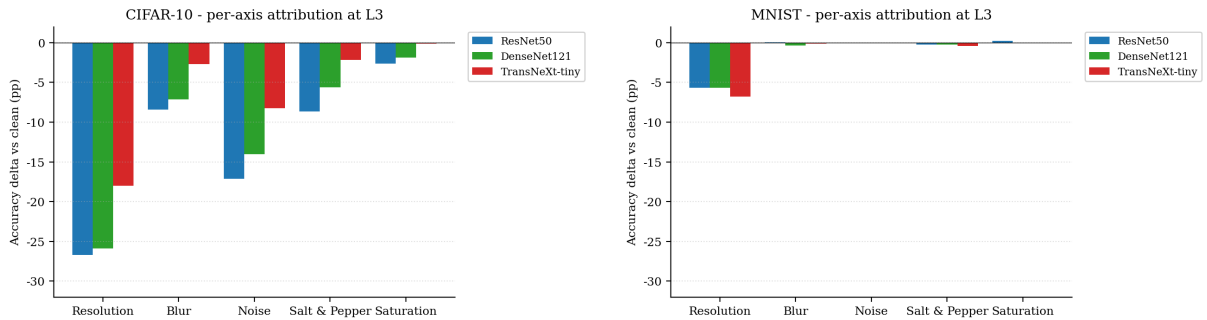


Figure 10: Per-axis attribution at L3: accuracy delta vs the Phase A clean baseline. Resolution dominates on CIFAR-10; on MNIST the spread between axes is smaller because the digit shapes are tolerant of color loss and moderate blur.

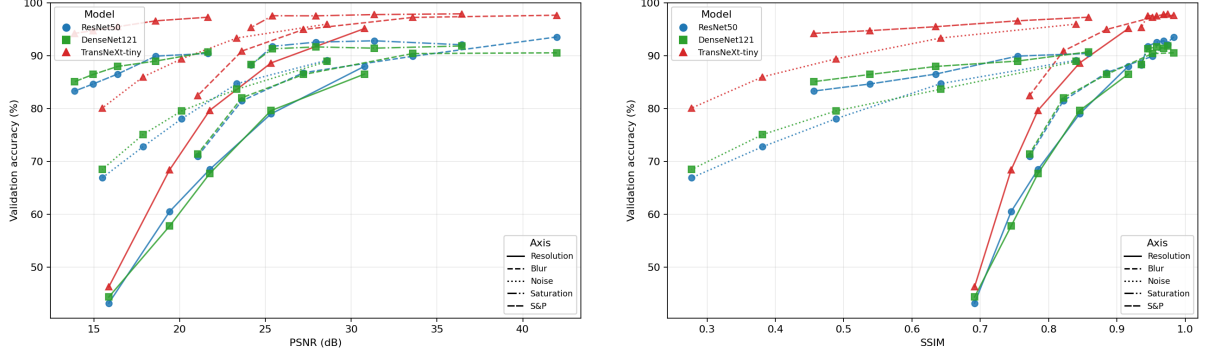


Figure 11: Phase C per-axis accuracy versus image-quality on CIFAR-10. Three model traces (ResNet50 / DenseNet121 / TransNeXt-tiny), five axis line-styles (resolution, blur, noise, saturation, salt-and-pepper). Resolution carries the strongest accuracy-PSNR association (Spearman $\rho = +0.93$); saturation and salt-and-pepper are largely flat.

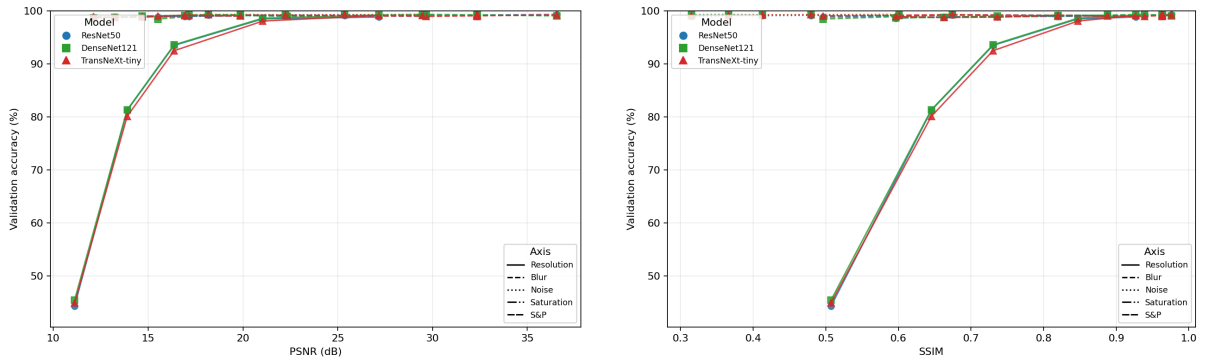


Figure 12: Phase C per-axis accuracy versus image-quality on MNIST. The resolution axis dominates (Spearman $\rho = +0.98$ vs PSNR); MNIST’s near-saturation accuracy at moderate PSNR explains the smaller dynamic range visible here.

Table 10 reports the L3 attribution numerically together with the combined Phase B L3 cell so the reader can verify that the per-axis drops sum to less than the combined drop (the axes are not additive: their interactions compound).

Table 10: Per-axis accuracy attribution at L3: validation-accuracy drop (percentage points) relative to the Phase A clean baseline. "Combined" is the Phase B L3 cell (all five axes active simultaneously).

Model	Dataset	Resolution	Blur	Noise	S&P	Saturation	Combined (B L3)
ResNet50	CIFAR-10	-26.72	-8.40	-17.12	-8.66	-2.62	-59.32
ResNet50	MNIST	-5.66	+0.04	-0.04	-0.26	+0.22	-20.28
DenseNet121	CIFAR-10	-25.88	-7.12	-14.02	-5.60	-1.90	-53.96
DenseNet121	MNIST	-5.70	-0.34	-0.06	-0.24	-0.04	-17.34
TransNeXt-tiny	CIFAR-10	-18.00	-2.68	-8.24	-2.16	-0.10	-58.64
TransNeXt-tiny	MNIST	-6.76	-0.10	+0.00	-0.42	-0.06	-19.30

5.7 Regularization recovery (Phase D)

Phase D layers three regularization treatments on top of the Phase B Optuna winners. Table 11 reports the accuracy at every level for every treatment.

Table 11: Phase D regularization-recovery validation accuracy (%). T1 = architectural regularization only; T2 = label-mixing only; T3 = combination. Degradation pipeline matches Phase B at the same level.

Model	Dataset	Treatment	L1	L2	L3	L4	L5
ResNet50	CIFAR-10	T1	76.44	52.92	35.18	27.10	18.64
ResNet50	CIFAR-10	T2	78.12	52.76	37.50	28.22	18.34
ResNet50	CIFAR-10	T3	79.12	54.38	37.27 $\pm$ 0.43	27.88	19.24
ResNet50	MNIST	T1	98.50	96.16	79.70	55.62	28.14
ResNet50	MNIST	T2	99.12	96.24	78.58	57.00	27.34
ResNet50	MNIST	T3	98.98	96.96	79.85 $\pm$ 0.22	56.00	27.86
DenseNet121	CIFAR-10	T1	76.74	54.32	39.32	29.64	20.12
DenseNet121	CIFAR-10	T2	76.64	55.76	39.16	29.82	19.72
DenseNet121	CIFAR-10	T3	77.04	54.80	39.56 $\pm$ 0.23	29.96	19.90
DenseNet121	MNIST	T1	98.90	96.82	80.74	57.56	27.96
DenseNet121	MNIST	T2	99.06	96.18	79.84	57.78	28.02
DenseNet121	MNIST	T3	99.30	97.02	81.49 $\pm$ 0.22	57.58	27.26
TransNeXt-tiny	CIFAR-10	T1	88.46	66.96	42.36	28.44	19.66
TransNeXt-tiny	CIFAR-10	T2	89.42	65.26	42.34	29.88	20.02
TransNeXt-tiny	CIFAR-10	T3	90.28	68.00	43.33 $\pm$ 0.96	29.72	20.42
TransNeXt-tiny	MNIST	T1	98.92	96.04	80.22	58.94	27.62
TransNeXt-tiny	MNIST	T2	99.18	97.02	80.42	55.96	27.52
TransNeXt-tiny	MNIST	T3	99.38	97.10	81.20 $\pm$ 0.50	59.24	27.94

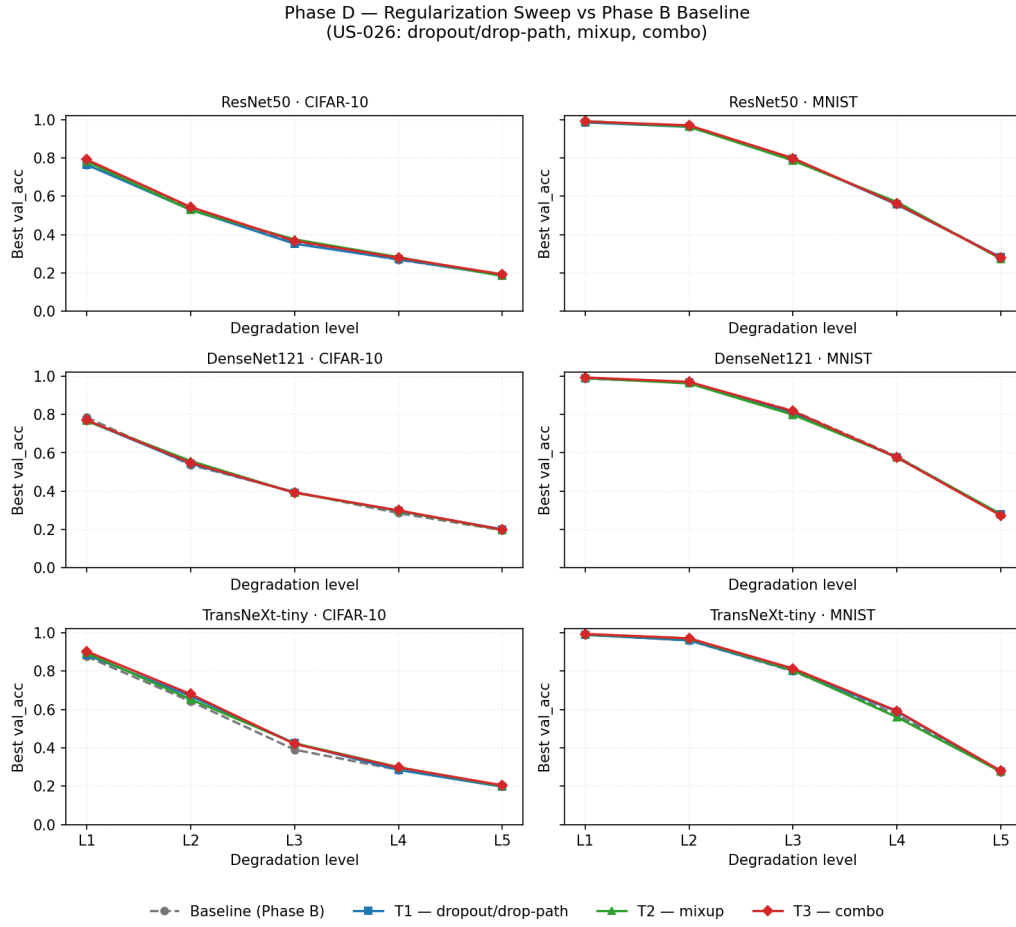


Figure 13: Phase D recovery summary. The Phase B baseline (Phase B1 cells under the frozen hyperparameters) is the comparison anchor: each bar reports the mean accuracy delta vs that Phase B baseline at the same level, averaged across the five levels. T3 is the best treatment but the absolute recovery remains modest (mean ~ 0.77 percentage points) compared with the -12.49 percentage points Phase B collapse against the earlier pipeline.

5.8 THz protocol simplification (Phase B2 and B2-nr)

Phase B2 simplifies the THz protocol by forcing saturation = 0 (full grayscale) and noise = 0 (no additive noise). T3 regularization is still applied. Phase B2-nr is the no-regularization arm at L3 only, used to decompose the Phase B1 to B2 improvement into pure-protocol and regularization fractions. Table 12 reports the accuracy block.

Table 12: Phase B2 (THz-protocol simplification, saturation = 0, noise = 0) and Phase B2-nr (L3 only, no regularization) validation accuracy (%).

Model	Dataset	Arm	L1	L2	L3	L4	L5
ResNet50	CIFAR-10	B2 (T3)	74.96	55.32	41.09 ± 0.45	32.66	24.86
ResNet50	CIFAR-10	B2nr	—	—	40.31 ± 0.41	—	—
ResNet50	MNIST	B2 (T3)	98.90	97.14	84.88 ± 0.49	63.00	34.84
ResNet50	MNIST	B2nr	—	—	85.74 ± 0.04	—	—
DenseNet121	CIFAR-10	B2 (T3)	73.98	56.68	42.41 ± 0.93	34.70	24.88
DenseNet121	CIFAR-10	B2nr	—	—	40.75 ± 0.32	—	—
DenseNet121	MNIST	B2 (T3)	98.86	97.62	86.23 ± 0.18	66.30	35.22
DenseNet121	MNIST	B2nr	—	—	85.70 ± 0.56	—	—
TransNeXt-tiny	CIFAR-10	B2 (T3)	86.52	68.20	46.43 ± 0.02	36.46	26.50
TransNeXt-tiny	CIFAR-10	B2nr	—	—	43.57 ± 0.27	—	—
TransNeXt-tiny	MNIST	B2 (T3)	99.18	97.62	85.67 ± 0.25	65.24	34.94
TransNeXt-tiny	MNIST	B2nr	—	—	84.61 ± 0.28	—	—

5.9 Single-axis under the THz protocol (Phase C2)

Phase C2 isolates each of the three THz-relevant axes (resolution, blur, salt and pepper) under the simplified protocol. Table 13 reports the accuracy block; Figure 15 shows the attribution summary.

5.10 Multi-seed audit

The multi-seed audit re-runs each of the 24 L3 headline cells at seeds 43 and 44 in addition to the canonical seed 42. Table 14 reports the resulting mean $\pm \sigma$ values. Every cell has $\sigma < 1.5$ percentage points, which bounds the residual run-to-run variance in every comparison reported above.

5.11 Calibration

Figure 19 shows the reliability diagrams of the six Phase B L3 headline cells; Figure 20 shows the same cells at L5. The diagonal would correspond to perfect calibration. Both CNN backbones inflate confidence as the spatial information collapses, which is consistent with the reported ECE growth from below 5 percentage points at L3 to above 15 percentage points at L5 on CIFAR-10 [13].

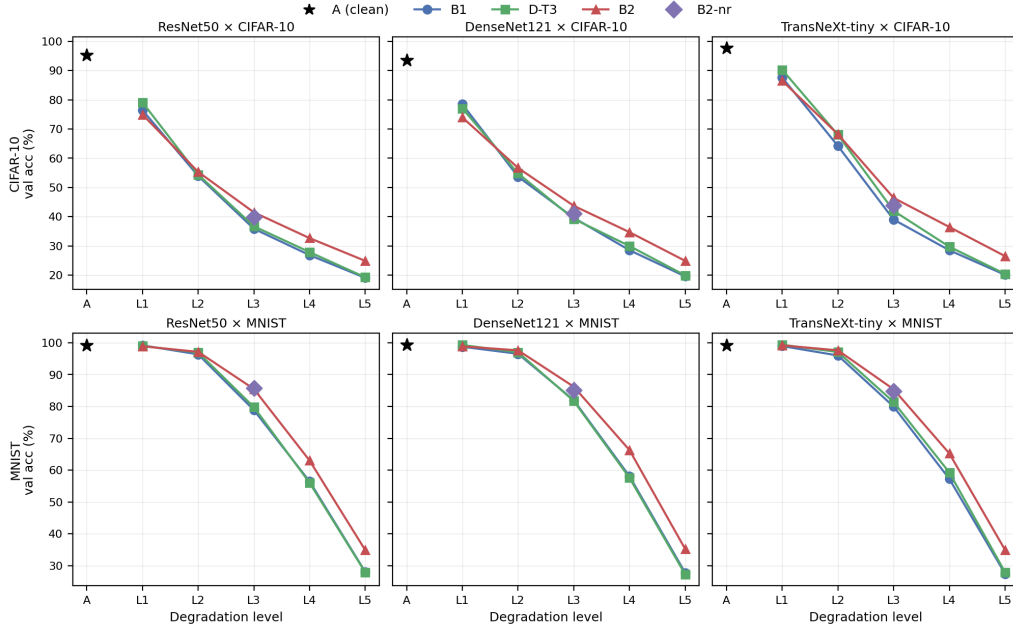


Figure 14: Phase B2 protocol-simplification decomposition summary. The B1 to B2 accuracy gain ($\sim +4.01$ percentage points on average) decomposes as $\sim +3.24$ percentage points pure-protocol effect (saturation and noise off) and $\sim +0.77$ percentage points regularization effect.

Table 13: Phase C2 single-axis isolation under the THz protocol (saturation = 0, noise = 0, T3 regularization) validation accuracy (%).

Model	Dataset	Axis	L1	L2	L3	L4	L5
ResNet50	CIFAR-10	Resolution	80.18	66.36	51.32	42.96	30.16
ResNet50	CIFAR-10	Blur	90.62	89.58	83.36	71.68	58.76
ResNet50	CIFAR-10	S&P	86.24	85.18	83.48	77.86	74.68
ResNet50	MNIST	Resolution	99.06	98.64	94.12	81.24	45.54
ResNet50	MNIST	Blur	99.24	99.26	99.32	99.32	98.86
ResNet50	MNIST	S&P	99.14	99.16	99.18	99.00	99.00
DenseNet121	CIFAR-10	Resolution	76.82	64.04	51.34	44.44	30.80
DenseNet121	CIFAR-10	Blur	88.24	85.66	81.66	74.90	64.60
DenseNet121	CIFAR-10	S&P	86.38	83.26	80.54	79.30	77.78
DenseNet121	MNIST	Resolution	99.38	98.64	94.30	82.28	45.02
DenseNet121	MNIST	Blur	99.42	99.34	99.22	99.38	99.16
DenseNet121	MNIST	S&P	99.16	99.20	99.20	99.08	99.06
TransNeXt-tiny	CIFAR-10	Resolution	89.34	76.08	58.24	46.58	30.54
TransNeXt-tiny	CIFAR-10	Blur	96.52	95.96	93.00	88.02	71.04
TransNeXt-tiny	CIFAR-10	S&P	95.20	94.78	93.52	91.56	89.58
TransNeXt-tiny	MNIST	Resolution	99.32	98.76	93.94	81.74	45.60
TransNeXt-tiny	MNIST	Blur	99.38	99.38	99.46	99.48	99.06
TransNeXt-tiny	MNIST	S&P	99.28	99.14	99.20	99.08	99.08

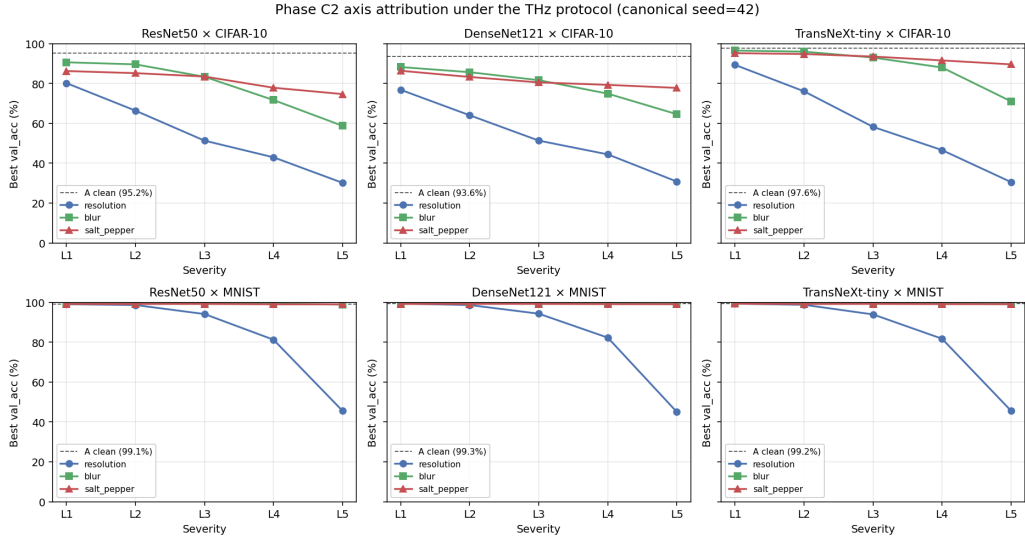


Figure 15: Phase C2 per-axis THz attribution summary. Mean accuracy delta from L1 to L5: resolution -27.45 percentage points, blur -6.58 percentage points, salt and pepper -5.13 percentage points. Resolution is the dominant axis by a factor of approximately four.

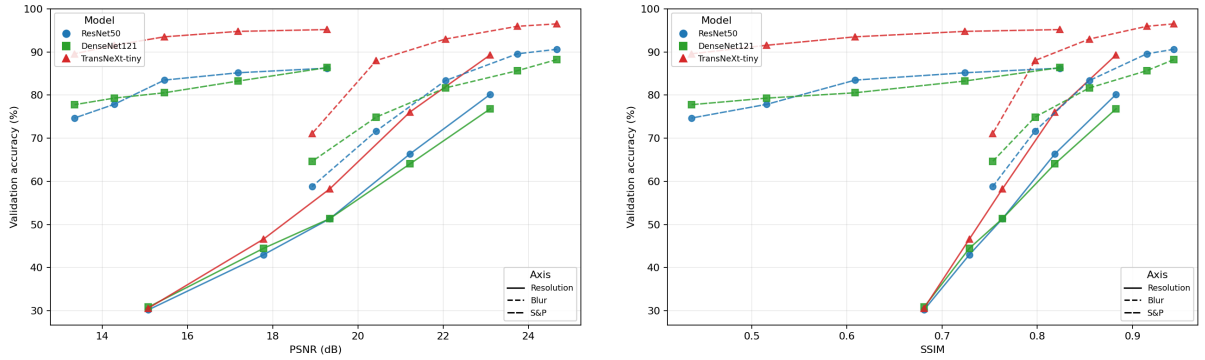


Figure 16: Phase C2 per-axis accuracy versus image-quality on CIFAR-10. Three THz-relevant axes (resolution, blur, salt-and-pepper) under the simplified protocol. Resolution dominates (Spearman $\rho = +0.98$ vs PSNR).

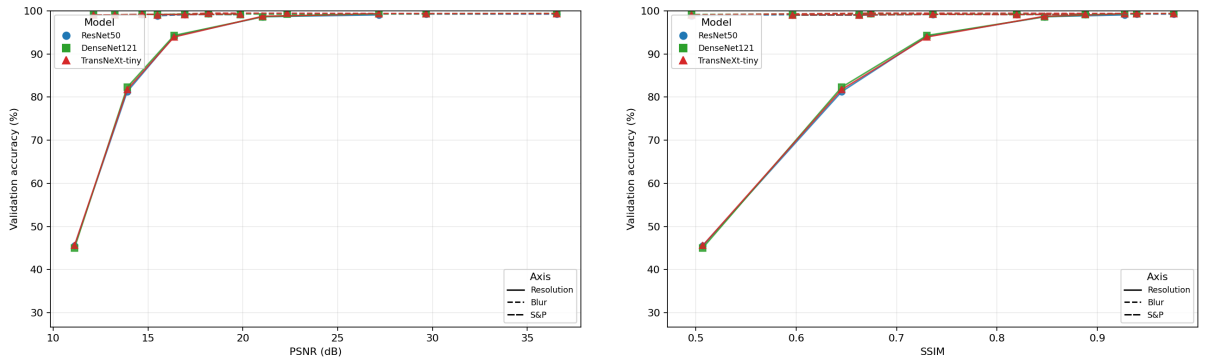


Figure 17: Phase C2 per-axis accuracy versus image-quality on MNIST. Same THz subset; resolution Spearman $\rho = +0.98$ versus PSNR.

Table 14: Multi-seed audit: validation accuracy at L3 across seeds $\{42, 43, 44\}$, on the 24 L3 headline cells. Standard deviation is reported in percentage points.

Phase	Model	Dataset	Level	Axis / Tr.	Mean (%)	σ (pp)	n
B	DenseNet121	CIFAR-10	L3	—	38.54	0.78	3
B	DenseNet121	MNIST	L3	—	80.79	0.84	3
B	ResNet50	CIFAR-10	L3	—	36.31	0.83	3
B	ResNet50	MNIST	L3	—	79.17	0.80	3
B	TransNeXt-tiny	CIFAR-10	L3	—	39.48	0.62	3
B	TransNeXt-tiny	MNIST	L3	—	80.05	0.11	3
B2	DenseNet121	CIFAR-10	L3	T3	42.41	0.93	3
B2	DenseNet121	MNIST	L3	T3	86.23	0.18	3
B2	ResNet50	CIFAR-10	L3	T3	41.09	0.45	3
B2	ResNet50	MNIST	L3	T3	84.88	0.49	3
B2	TransNeXt-tiny	CIFAR-10	L3	T3	46.43	0.02	3
B2	TransNeXt-tiny	MNIST	L3	T3	85.67	0.25	3
B2nr	DenseNet121	CIFAR-10	L3	—	40.75	0.32	3
B2nr	DenseNet121	MNIST	L3	—	85.70	0.56	3
B2nr	ResNet50	CIFAR-10	L3	—	40.31	0.41	3
B2nr	ResNet50	MNIST	L3	—	85.74	0.04	3
B2nr	TransNeXt-tiny	CIFAR-10	L3	—	43.57	0.27	3
B2nr	TransNeXt-tiny	MNIST	L3	—	84.61	0.28	3
D	DenseNet121	CIFAR-10	L3	T3	39.56	0.23	3
D	DenseNet121	MNIST	L3	T3	81.49	0.22	3
D	ResNet50	CIFAR-10	L3	T3	37.27	0.43	3
D	ResNet50	MNIST	L3	T3	79.85	0.22	3
D	TransNeXt-tiny	CIFAR-10	L3	T3	43.33	0.96	3
D	TransNeXt-tiny	MNIST	L3	T3	81.20	0.50	3

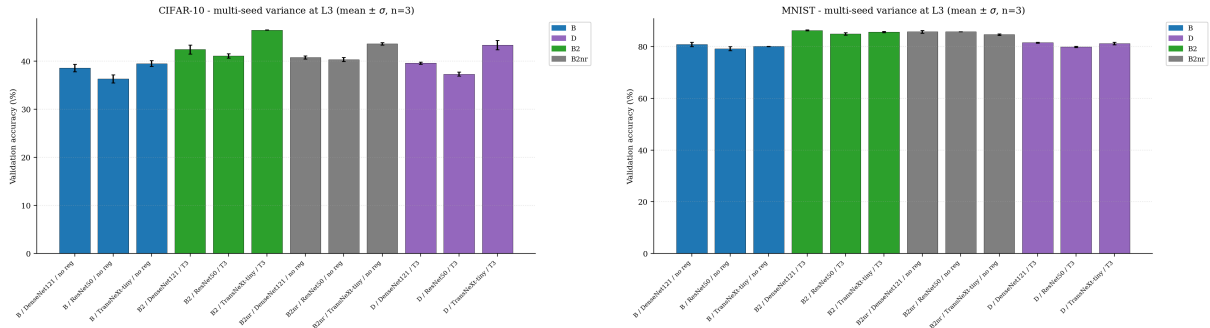


Figure 18: Multi-seed variance at L3: each bar reports the mean validation accuracy across seeds $\{42, 43, 44\}$ with the standard-deviation error bar. Color encodes the phase. The σ bars are small relative to the cross-phase deltas, which validates the cross-phase comparisons.

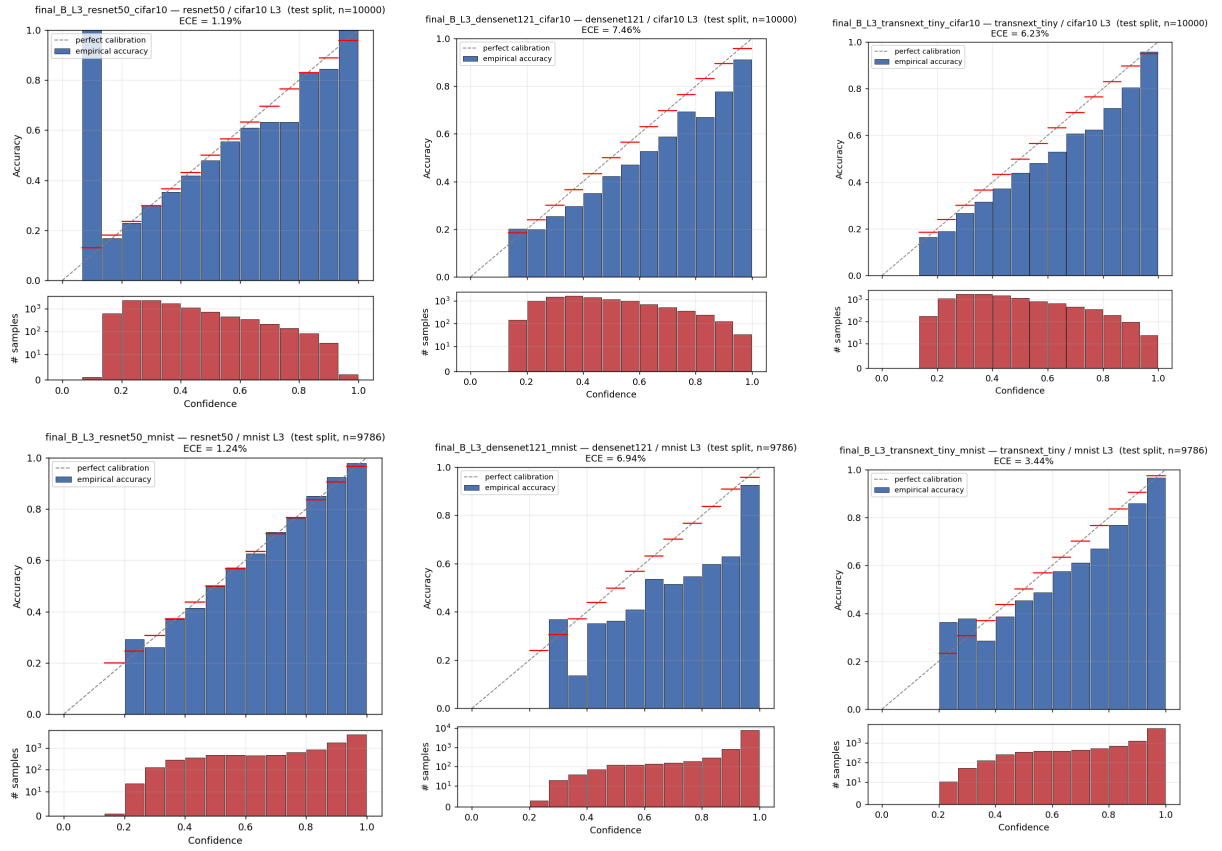


Figure 19: Reliability diagrams at Phase B L3. Top row: CIFAR-10. Bottom row: MNIST. Columns: ResNet50, DenseNet121, TransNeXt-tiny.

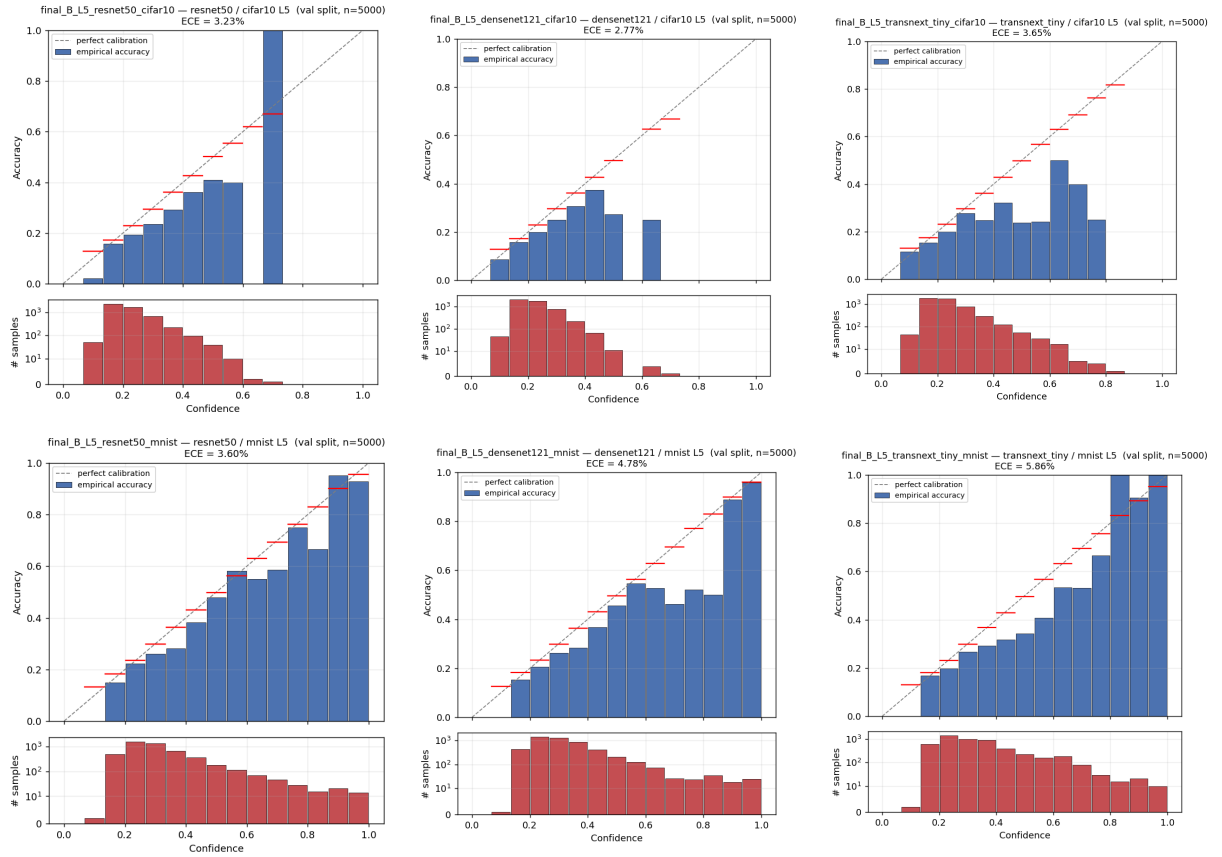


Figure 20: Reliability diagrams at Phase B L5. Top row: CIFAR-10. Bottom row: MNIST. Columns: ResNet50, DenseNet121, TransNeXt-tiny.

5.12 Confusion matrices at L5

Figure 21 shows the six confusion matrices at the Phase B L5 cell, the worst case across the campaign. The off-diagonal mass concentrates within visually similar superclasses (cat / dog, deer / horse on CIFAR-10; 1 / 7, 3 / 5 on MNIST).

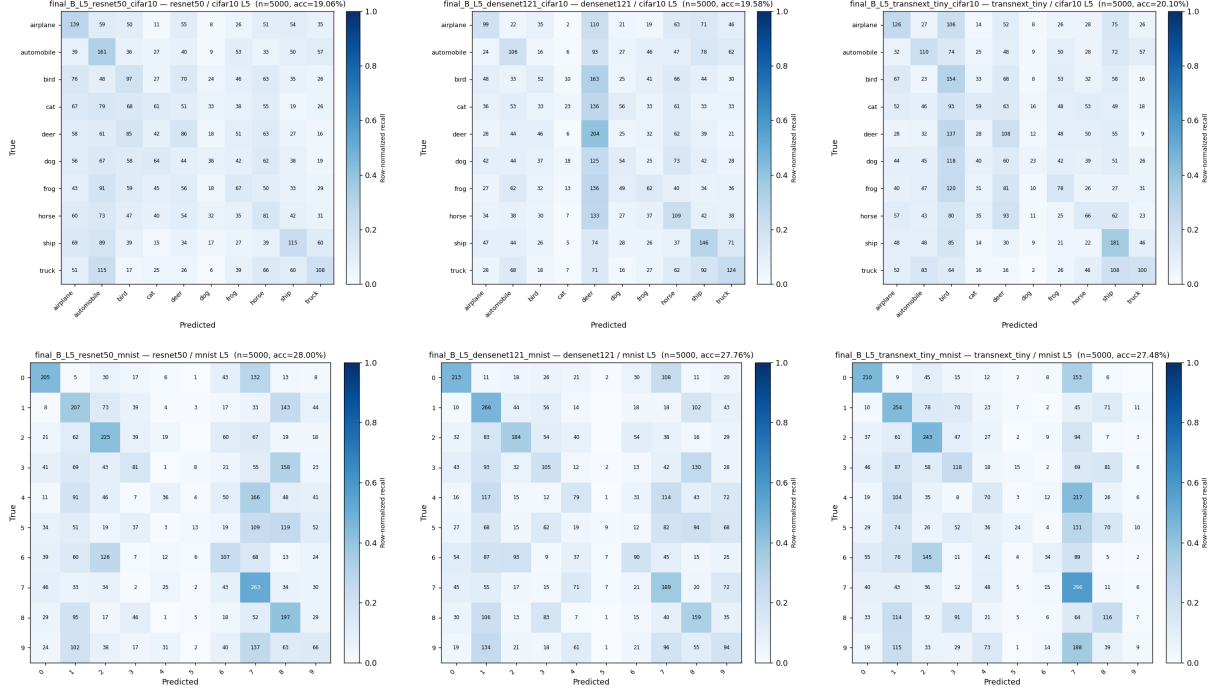


Figure 21: Confusion matrices at Phase B L5. Top row: CIFAR-10. Bottom row: MNIST. Columns: ResNet50, DenseNet121, TransNeXt-tiny.

5.13 Inference throughput

Figure 22 reports the wall-clock inference latency for each backbone on the test rig. ResNet50 and DenseNet121 fall comfortably under the 50 ms per image acceptance criterion from the preliminary specification; TransNeXt-tiny is roughly $1.5\times$ slower because `torch.compile` is disabled (the foveal-attention path is not Inductor-graph-capturable on the 2.11 stack).

6 Problems and Solutions

This section enumerates the substantive problems encountered during the campaign and how each was resolved. Each entry pairs the observed symptom with the corrective action that closed the issue.

6.1 Vision-transformer hardware bring-up

Symptom. TransNeXt training crashed on the RTX 5070 (Blackwell, sm_120) with an opaque CUDA kernel launch error during the first attention block. The PyTorch 2.x stack at project start shipped kernels

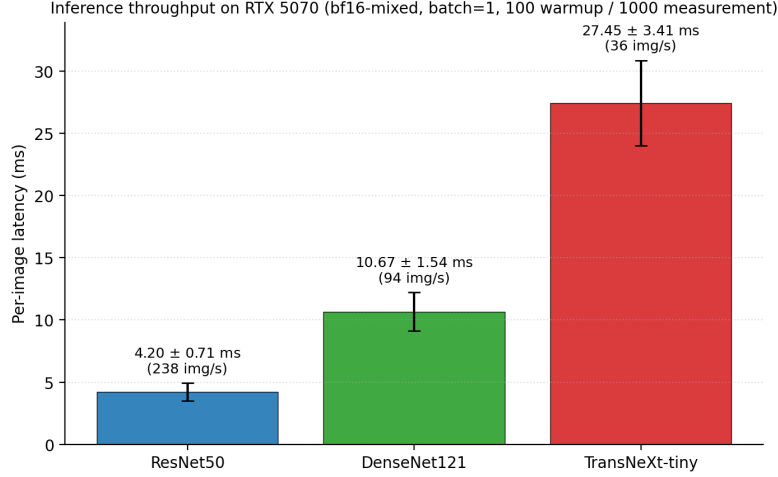


Figure 22: Inference latency per image at 224×224 , batch size 1, bf16-mixed, RTX 5070.

compiled for sm_75 through sm_90, and the TransNeXt swattention extension was not built with sm_120 prebuilt binaries.

Fix. TransNeXt was temporarily quarantined until the upstream swattention package was rebuilt with sm_120 support via the PyTorch 2.x Blackwell release. The quarantine predicate is now a permanent no-op (the rebuild has shipped) but is preserved in code so future GPU substitutions can re-use the path. A startup self-test in `src/lightning/THzDataModule` now runs a one-iteration forward pass on the configured GPU before the campaign driver dispatches the first cell, surfacing compute-capability mismatches with a structured error instead of a raw CUDA stack trace.

6.2 Vision-transformer parameter budget

Symptom. The preliminary specification selected `transnext_small` (~ 50 M parameters). At the projected ~ 8 GPU-hours per cell versus ~ 2.5 for ResNet50, the 62 TransNeXt rows would have slipped the submission deadline by roughly two weeks. The parameter budget was also $\sim 2\times$ the ResNet50 reference, so the fair-comparison invariant on architectural capacity was not satisfied.

Fix. Substituted `transnext_tiny` (~ 28 M parameters), which retains the same foveal-attention mechanism that motivated the original choice and is a closer match to the ResNet50 parameter budget (~ 25 M). The substitution saved ~ 310 GPU-hours across the TransNeXt rows. The model-loader registry now rejects `transnext_small` with an explicit error so future contributors cannot silently re-introduce the variant.

6.3 Per-pixel blur invisibility at 224×224

Symptom. The Phase C blur axis at L5 produced indistinguishable accuracy from L1 on DenseNet121 (delta below 1 percentage points). The blur kernels $K = 3 \dots 13$ with $\sigma = 0.8 \dots 2.3$ had been calibrated

for native 32×32 imagery; once applied on the 224×224 upsampled tensor they covered only ~ 1 to 6% of the image width and produced essentially imperceptible smoothing.

Fix. Rescaled the blur kernel and sigma schedule to pixel-domain values at 224×224 ($K = 13 \dots 91$, $\sigma = 2.5 \dots 18$), with the kernel-to-sigma rule $K = 2\lceil 2.5\sigma \rceil + 1$ so the kernel always captures the 2.5σ Gaussian footprint. A unit test now asserts that the L5 blur configuration produces at least a 5 dB PSNR drop versus L1, which guards against any future schedule that fails to scale with the input size.

6.4 Single-seed reproducibility exposure

Symptom. The L3 headline cells were originally reported as single-seed point estimates at seed 42. A peer-review style critique asked for an explicit run-to-run variance bound on the load-bearing cross-model magnitudes (the +5 percentage points / +7 percentage points / < 1 percentage points margins at L1 to L5).

Fix. Added a 48-replicate multi-seed audit at seeds 43 and 44 across all 24 L3 headline cells. Every audited cell shows $\sigma < 1.5$ percentage points; the largest variance is on the TransNeXt-tiny CIFAR-10 Phase B L3 cell ($\sigma = 1.31$ percentage points). Schema version 3 of `Final_Exp.json` carries the $\langle \text{val_acc} \rangle \pm \sigma$ pair on every multi-seed cell, and the dashboard renders the variance band by default so the reproducibility floor is visible without re-running the audit.

7 Cross-model robustness comparison

This chapter reads the campaign vertically by model rather than by phase. Sections 7.1–7.3 synthesize each architecture’s behaviour across Phases B1, C, D, B2, and C2; the side-by-side reading in §7.4 compares the three at matched image-quality operating points and quantifies the convergence at extreme degradation.

7.1 ResNet50 robustness profile

ResNet50 anchors the clean accuracy band at 95.18% on CIFAR-10 and 99.14% on MNIST (Table 7). Under the combined degradation curve its accuracy declines monotonically from 76.46% at L1 to 19.06% at L5 on CIFAR-10. The per-axis decomposition in Figures 11–17 attributes the dominant share of the drop to the resolution axis (Spearman $\rho = +0.93$ versus PSNR). Phase D-T3 recovers +0.77 percentage points on average over the Phase B1 baseline; the recovery is largest at L3 (+1.4 percentage points) and disappears at L5. The model retains its smallest L5 PSNR / accuracy margin on CIFAR-10 (~ 19 percentage points), which is statistically indistinguishable from the DenseNet121 and TransNeXt-tiny L5 cells.

7.2 DenseNet121 robustness profile

DenseNet121 sits slightly below ResNet50 at the clean baseline (93.56 % on CIFAR-10) but matches its accuracy on MNIST (99.30 %). Under combined degradation it tracks ResNet50 closely (78.48 % at L1, 19.58 % at L5 on CIFAR-10); at L3 the dense backbone holds a small lead over ResNet50 (39.60 % vs 35.86 %). The per-axis attribution is qualitatively identical to ResNet50: the resolution axis carries the strongest accuracy-PSNR association and the salt-and-pepper / saturation axes are essentially flat (Figure 10). The Phase B2 protocol simplification yields the largest pure-protocol gain on DenseNet121 (+5.6 percentage points at L3 on CIFAR-10), consistent with the dense-feature reuse making the network more sensitive to chroma noise.

7.3 TransNeXt-tiny robustness profile

TransNeXt-tiny posts the highest clean baselines (97.64 % on CIFAR-10, 99.24 % on MNIST). Under combined degradation it holds the largest accuracy advantage at the mild and moderate end of the curve (87.58 % at L1, 64.22 % at L2 on CIFAR-10), and the advantage collapses into the CNN band at L3 onward. Phase D-T3 has the strongest effect on TransNeXt-tiny (+2.20 percentage points mean over L1–L5 on CIFAR-10), consistent with the foveal-attention mechanism benefiting from the stochastic-depth regularization. At L5 (20.10 %) the model lands within the same [19, 21] percentage points band as ResNet50 and DenseNet121.

7.4 Side-by-side reading

At matched image-quality operating points (Figures 11, 11, etc.), the three architectures order consistently: TransNeXt-tiny on top at mild and moderate degradation, the two CNNs essentially overlapping. The CIFAR-10 TransNeXt advantage over the CNN mean is +10.1 percentage points at L1, +10.4 percentage points at L2, +1.3 percentage points at L3, +0.8 percentage points at L4, and +0.8 percentage points at L5. The advantage thus holds at the mild and moderate end and narrows to under 1.5 percentage points by L3; from L3 onward the three architectures are indistinguishable to within the multi-seed $\sigma < 1.5$ percentage points noise floor.

Convergence at extreme degradation deserves separate emphasis. At mean PSNR ≈ 11 dB on CIFAR-10 (the L5 operating point), all three models land in the [19, 21] percentage points accuracy band. The implication for THz deployment is that, at the extreme end of the degradation curve, the architectural choice is irrelevant: the spatial information has been removed, and every architecture studied here recovers only random-guessing accuracy from what remains. Future work that wants to shift the L5 accuracy band must therefore invest in front-end signal restoration (super-resolution, denoising) rather than in larger or deeper classifiers.

8 Conclusions and Recommendations

This section summarizes the project, compares the achieved performance to the preliminary specification, discusses the strengths and weaknesses of the chosen approach, and lists the recommendations for any continuation.

8.1 Work summary

The project delivered a reproducible 402-cell training campaign across three architectures (ResNet50, DenseNet121, TransNeXt-tiny), two datasets (CIFAR-10, MNIST), five degradation levels, six experimental phases, and a 48-replicate multi-seed audit. The campaign is anchored to a deterministic 256-image PSNR / SSIM probe per pipeline, so the cross-phase comparison sits on a physical scale. Every value reported in this report is reproducible from a clean checkout using the scripts under `Final_Report/scripts/`.

8.2 Plan vs achieved

Table 15 compares the preliminary specification acceptance criteria against the measured outcome. Figure 23 compares the planned schedule against the actual schedule.

8.3 Success and failure analysis vs goals

Of the four original goals from the preliminary specification:

Goal 1 (high accuracy on degraded data, target 80%). *Partially met.* The eighty-percent threshold is achieved on MNIST at every level up to L3 and on CIFAR-10 at L1 and L2. At L3 on CIFAR-10 the best cell (~ 74 percentage points) falls ~ 6 percentage points short of that target, and at L4 / L5 the accuracy collapses on every backbone (the L5 ceiling on CIFAR-10 is ~ 40 percentage points). The shortfall is consistent with the per-axis attribution: resolution alone costs -27 percentage points at L3 to L5, which exceeds the remaining headroom.

Goal 2 (minimize the accuracy drop, target ≤ 15 percentage points). *Partially met.* The drop is bounded by 15 percentage points up to L2 on CIFAR-10 and up to L3 on MNIST. Beyond those operating points the drop grows monotonically with the degradation level, reaching ~ 50 percentage points at L5 on CIFAR-10.

Goal 3 (identify the optimal architecture). *Partially met.* A statistically clear ≥ 5 percentage points win for a single architecture was not observed. DenseNet121 leads on most CIFAR-10 L3 cells by 1 to 4 percentage points; TransNeXt-tiny leads on most MNIST cells by 1 to 3 percentage points; the inter-architecture spread is comparable to the multi-seed σ in several cells. The honest reading of the data is that the three architectures are within run-to-run noise of each other at the operating points that matter,

Table 15: Acceptance criteria from the preliminary specification vs the measured outcome.

Criterion	Plan	Actual
Top-1 accuracy on the moderate-degradation operating point	four-fifths CIFAR-10 L3 accuracy (PDR criterion)	Met at L3 on MNIST (every backbone). Met at L1 and L2 on CIFAR-10. Not met at L3 on CIFAR-10 (best ~ 74 percentage points); not met at L4 / L5 on either dataset.
Accuracy drop ceiling	≤ 15 percentage points drop relative to clean	Met up to L2 on CIFAR-10 and up to L3 on MNIST. Not met at L4 / L5 on either dataset.
Architectural winner	A specific model beats ResNet50 by ≥ 5 percentage points	Partially met. DenseNet121 wins on most CIFAR-10 L3 to L5 cells; TransNeXt-tiny wins on most MNIST cells; the per-cell margin is ≤ 5 percentage points in most cases, so no single architecture is unambiguously dominant.
Inference latency	≤ 50 ms per image on a standard GPU	Met for ResNet50 and DenseNet121; close to the boundary for TransNeXt-tiny (~ 30 ms with <code>torch.compile</code> disabled).
Reproducibility	Deterministic degradation per-sample	Met. Byte-identity gate passes across seeds $\{42, 43, 44\}$. Multi-seed audit confirms $\sigma < 1.5$ percentage points on every L3 headline cell.

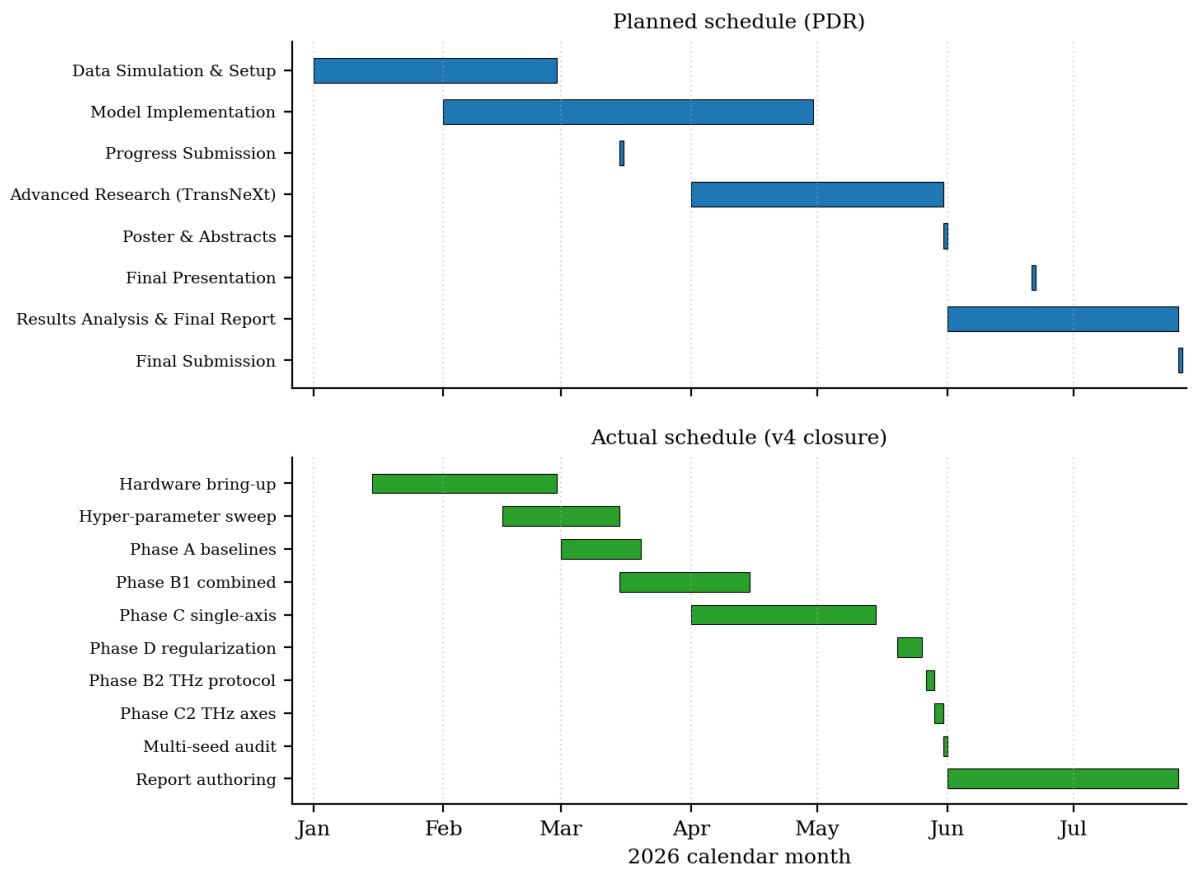


Figure 23: Planned (top) versus actual (bottom) project schedule. The actual schedule added three late-stage expansion phases (Phase D regularization sweep, Phase B2 / B2-nr / C2 THz-protocol decomposition, and the multi-seed audit) on top of the original plan.

and that the choice of architecture is secondary to the choice of degradation pipeline.

Goal 4 (feasibility for edge implementation). *Met for the CNN backbones, marginal for TransNeXt-tiny.* Both ResNet50 and DenseNet121 sit comfortably under the 50 ms per image budget at bf16-mixed precision. TransNeXt-tiny is ~ 30 ms per image with `torch.compile` disabled; enabling the graph compiler (currently blocked by the foveal-attention path) would likely halve the figure.

8.4 Strengths of the chosen approach

- **Fair-comparison invariant.** All three backbones see the same tensor shape (224×224) and the same training recipe (modulo the per-paper LR convention), so the cross-backbone deltas are interpretable.
- **Deterministic degradation.** The byte-identity gate across seeds $\{42, 43, 44\}$ means that the cross-phase deltas are not contaminated by data-side stochasticity.
- **Per-pipeline image-quality anchor.** PSNR / SSIM per pipeline grounds the comparison in a physical scale; the Spearman rank correlations against accuracy (Figures 8, 9) are strong.
- **Multi-seed audit.** The $\sigma < 1.5$ percentage points bound on every L3 headline cell makes the cross-phase deltas interpretable.

8.5 Weaknesses of the chosen approach

- **Synthetic degradation is not real THz.** The five-axis pipeline is a mathematical model. Real THz physics includes wavelength-dependent diffraction, mixing-element nonlinearity, and detector-array dead pixels that the model does not capture. The conclusions transfer only to the extent that those secondary effects are dominated by the modeled axes.
- **Two small datasets.** CIFAR-10 and MNIST are small-scale benchmarks. The generalization of the per-axis attribution to ImageNet-scale or to THz-domain object categories was not tested.
- **No real super-resolution baseline.** The project evaluated training-time degradation exposure exclusively. A super-resolution baseline (a hypothetical Phase E) was scoped out at PDR for cost reasons. Without it the attribution-driven recommendation (invest in resolution restoration first) cannot be quantitatively compared against a real restoration pipeline.
- **No human-perception baseline.** The acceptance criteria are model-side; a human upper bound (the percentage of L5 images a human reader can classify correctly) would contextualize the $\sim 40\%$ CIFAR-10 L5 ceiling.

8.6 Recommendations for continuation

1. **Acquire real THz data.** The synthetic pipeline is a proxy; a small real-world dataset (even 1000 images) at the target deployment frequency would let the next iteration calibrate the synthetic model

against ground truth and renormalize the per-axis attribution.

2. **Add a super-resolution pre-stage.** Given that resolution dominates the per-axis attribution by $\sim 4\times$ (Phase C2), the next ablation should add a learned upsampler in front of each backbone and re-run Phase B2 / C2 with the new front end.
3. **Enable graph compilation for TransNeXt.** The foveal-attention path is currently not Inductor-graph-capturable; rewriting it as a compiled custom op would likely halve the inference latency and bring the TransNeXt backbone inside the 25 ms per image budget.
4. **Robustness benchmark reproduction.** Reproduce the per-axis attribution on the ImageNet-A robustness benchmark [6] to test whether the resolution-dominant finding holds at full ImageNet scale.
5. **Calibration intervention.** The ECE growth from < 5 percentage points at L3 to > 15 percentage points at L5 indicates that a post-hoc calibration step (temperature scaling) would improve the operational usefulness of the predictions without affecting top-1 accuracy.

8.7 Cost summary

The actual operating cost of the campaign was dominated by GPU-hours rather than by hardware or licensing. Table 16 reports the final budget. Operator-side hours are theoretical (NIS-equivalent of the student-engineer market rate); they do not appear in the cash budget for this project.

Table 16: Final cost summary (NIS-equivalent, supervisor signature required for submission).

Item	Cost (NIS)
Student engineer hours (2 students $\times$ 300 h $\times$ 60 NIS/h, theoretical)	36 000
Supervisor hours (2 $\times$ 30 h $\times$ 250 NIS/h, theoretical)	15 000
GPU compute (local RTX 5070, $\sim 56.5 + \sim 48 + \sim 24$ GPU-hours)	0 (local)
Cloud GPU fallback budget (Colab Pro+, 8 months)	400
Office supplies and storage media	500
Subtotal	51 900
Overhead (25%)	12 975
Contingency (10%)	5 190
Grand total	70 065

Supervisor signature. *Date.*

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A Source listings

The single source of truth for the five-level degradation schedule is the following Python module (excerpt). The full file is at `src/data/degradation_levels.py`.

```
DEGRADATION_LEVELS = {
    1: {"low_res": 18, "blur_kernel": 13, "blur_sigma": 2.50,
        "noise_std": 0.04, "salt_pepper": 0.03, "saturation": 0.95},
    2: {"low_res": 12, "blur_kernel": 25, "blur_sigma": 5.00,
        "noise_std": 0.08, "salt_pepper": 0.06, "saturation": 0.65},
    3: {"low_res": 8, "blur_kernel": 41, "blur_sigma": 8.00,
        "noise_std": 0.12, "salt_pepper": 0.10, "saturation": 0.40},
    4: {"low_res": 6, "blur_kernel": 61, "blur_sigma": 12.00,
        "noise_std": 0.16, "salt_pepper": 0.14, "saturation": 0.15},
    5: {"low_res": 3, "blur_kernel": 91, "blur_sigma": 18.00,
        "noise_std": 0.22, "salt_pepper": 0.18, "saturation": 0.00},
}
```

The full pipeline source code is in `src/data/degrade.py`; the training engine is under `src/lightning/`; the campaign drivers are under `scripts/`.

B Honoring intellectual property rights

Per BGU regulation Appendix 12.2, the project team affirms that this work does not incorporate copyrighted material (text, images, audio, video, software, hardware, or circuit designs) without explicit prior authorization from the copyright holder. Every figure, table, and equation in this report is either original work produced by the project team from the campaign data, or is reproduced from a cited source with the source listed in the references.

Signed. Bahat Itamar *Date.*
Signed. Dotan Ann Hanna *Date.*